Artifact-Gated Evaluation of a Causal FIR Module for Sequential Recommendation

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We test whether a small, identity-initialized causal finite-impulse-response (FIR) residual provides a robust modular gain in sequential recommendation. On three outcome-known Amazon Reviews 2023 category/split settings, learned FIR minus identity estimates are positive; on Musical_Instruments the effect is +0.002265, ordinary Welch 95% CI [0.001928, 0.002602]. The learned filter also exceeds an equal-parameter current-position-only placebo (+0.001941 [+0.001788, +0.002095]), but does not separate from a 16-parameter shared causal filter (-0.000081 [-0.000337, +0.000175]), limiting any claim of per-channel-tap necessity. A prospectively frozen same-investigator MovieLens 1M study is negative: learned minus identity +0.000000 [-0.000074, +0.000075] and learned minus pointwise +0.000035 [-0.000057, +0.000127]. Thus the parsimonious FIR arms support only conditional coefficient-count compression, not practical efficiency or cross-domain replication. Outcome-known same-evaluator AlphaFuse-style and WEARec studies improve comparator coverage; the AlphaFuse-style package remains above a separately frozen parser-default-normal SASRec-ID sensitivity by +0.005207 [+0.004779, +0.005635], but these are whole-package, unequal-model results below the paper's existing reference. Overall, the contribution is a narrow detachable implementation and an artifact-gated evaluation record, not a new architecture, general FIR benefit, SOTA result, or independent confirmation. A fail-closed graph preserves positive, null, deviated, and VOID outcomes under one reporting rule.

CCS Concepts: • **Information systems → Recommender systems**; *Personalization.*

Additional Key Words and Phrases: sequential recommendation, Amazon Reviews 2023, HSTU, causal FIR filtering, long-tail recommendation, text-augmented recommendation, reproducibility, pre-declaration

1 Introduction

Sequential recommendation models predict a user's next item from their interaction history. SASRec [Kang and McAuley 2018] and BERT4Rec [Sun et al. 2019] established Transformer-based baselines. Later systems add language-derived item representations [Ding et al. 2021; Hou et al. 2024, 2022; Li et al. 2023] or discrete semantic IDs [Rajput et al. 2023; Yang et al. 2024].

Amazon Reviews 2023 results span incompatible preprocessing variants, and superficially similar "5-core" labels hide different contracts. Our pipeline iteratively filters both users and items and then applies leave-last-out. LIGER reports user-and-item filtering, whereas TIGER reports user filtering only; both use Amazon Reviews 2014, not AR2023. Their holdout chronology is related to ours, but their datasets and filtering geometry are not interchangeable with our fixed AR2023 splits or with the 0-core reference pipeline.

Recommender evaluation is vulnerable to mistuned baselines, incompatible protocols, and irreproducible result selection [Ferrari Dacrema et al. 2021, 2019]. We therefore pair the method with a per-paper discipline: git-frozen protocols, a fail-closed graph that recomputes 201 reported cells across 25 claim families, environment-caveated comparator regeneration, and symmetric adjudication. Failures are retained. In particular, the original Office_Products protocol

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remains VOID after its own floor check exposed an environment mismatch, although a redesigned fresh-seed protocol later passed (§5.2; Appendix A.0).

Using an HSTU-style pure-PyTorch encoder based on Zhai et al. [2024], we ask whether one small causal filtering component survives matched controls and can be reused without category-specific tuning. Our contribution ledger is deliberately limited to three items:

(1) **A canonical causal FIR module.** We adapt earlier bidirectional frequency filters to a strictly left-causal, depth-wise residual that is identity-initialized and gradient-active under the all-position objective (§3). This is a modular component, not a new recommender architecture.

(2) **Internal evidence, a negative non-Amazon test, and a mechanism boundary.** Learned FIR–identity estimates are positive on three outcome-known Amazon category/split settings. A separately frozen Software attempt then produced +0.005062 [+0.004591, +0.005533], clearing its pre-declared +0.000500 reporting threshold with validation-only selection and sealed final evaluation. Because earlier V2 validation-output non-visibility is not independently established, V3 is classified outcome-known/exploratory under local same-user custody—not confirmation. On Musical_Instruments, learned taps do not separate from a shared causal filter, but they beat a parameter-matched current-position-only placebo by +0.001941 [+0.001788, +0.002095], while the placebo–identity interval spans zero. This discriminates learned FIR from that compound placebo; it does not isolate temporal access, establish unique necessity of per-channel taps or provide independent confirmation. A prospectively frozen MovieLens 1M study then failed to replicate learned FIR against identity or pointwise. Its shared/grouped/low-rank arms met a pre-declared noninferiority margin versus learned FIR only conditionally on that failed replication gate. Together these results prevent a general filtering or cross-domain claim. An official-code WEARec feasibility run under the same evaluator also scored below the existing full-model reference; it equalizes evaluation only, not architecture, loss, schedule, or tuning, and is neither independent confirmation nor SOTA evidence. A frozen study contrasting an AlphaFuse-style package with a zero-initialized upstream-class SASRec-ID control returned a positive whole-package contrast; the package also remained above a separately frozen parser-default-normal SASRec-ID sensitivity, but stayed below that full-model reference. These studies improve comparator coverage without isolating a component or supplying independent confirmation.

(3) **An auditable evaluation record.** Released protocols, artifacts, sidecars where available, comparator-regeneration code, and a strict rebuild preserve positive, null, deviated, and VOID outcomes under the same reporting rule (§8). The apparatus certifies reconstruction and protocol history; it does not strengthen the estimand.

TAPE, late fusion, sparse-warm redistribution, titrations, probe screens, and earlier Beauty results are supporting studies. All borrowed architectures and training recipes are attributed in §2.

2 Related Work and Attribution

We summarize prior work that our experiments directly build on. A complete attribution table for every component used in our experiments appears at the end of this section.

SASRec [Kang and McAuley 2018] is the base Transformer sequential recommender we re-implement. SASRec models a user's history with a causal-masked Transformer encoder and scores the next item via dot product with a learned item-embedding table.

BERT4Rec [Sun et al. 2019] adapts BERT-style bidirectional masked-item modeling for sequential recommendation. We implement BERT4Rec as one of our 20 ablation baselines on Beauty_and_PC.

SBERT / MiniLM [Reimers and Gurevych 2019; Wang et al. 2020] provide pre-trained sentence embeddings. We use `sentence-transformers/all-MiniLM-L6-v2` [Sentence-Transformers maintainers 2021] (384-d) as the default frozen item-text encoder.

UniSRec [Hou et al. 2022] is an early, widely-adopted text-based sequential recommender that uses a pre-trained language model to embed item text and learns a parametric adaptor to project these embeddings into the sequential model's hidden space; **ZESRec** [Ding et al. 2021] precedes it in representing items from natural-language descriptions with a pre-trained encoder for zero-shot sequential recommendation, and **RecFormer** [Li et al. 2023] subsequently studies language representations for sequential recommendation including low-resource and cold-start evaluation — we make no priority claim in this line. The SASRec-with-frozen-text-encoder pattern we use is a strict simplification of UniSRec.

BLaIR [Hou et al. 2024] (published as Hou et al. 2026) is a RoBERTa-base model continually pretrained on 10% of (item-metadata, review) pairs from Amazon Reviews 2023. We use `hyp1231/blair-roberta-base` (768-d) as an alternative frozen item-text encoder. Hou et al. also publish a reference implementation, **SASRecText**, that pairs SASRec with BLaIR via a 2-layer MLP adaptor `[768, 300, 64]` with Dropout(0.2) + ReLU; we adopt this adaptor design verbatim (see Appendix A.1).

TIGER [Rajput et al. 2023] reformulates sequential recommendation as autoregressive generation over discrete semantic IDs produced by a residual-quantized variational autoencoder over item-text embeddings. TIGER reports strong results on Amazon Reviews 5-core benchmarks.

LIGER [Yang et al. 2024] extends TIGER with a dense-retrieval refinement step and reports further gains.

Amazon Reviews 2023 [Hou et al. 2024] is the source dataset for the Amazon experiments. Our iterative **user-and-item 5-core leave-last-out** protocol (§3) shares chronological leave-out structure with TIGER and LIGER. LIGER states user-and-item filtering; the TIGER paper states user filtering, but we do not infer the exact geometry of its released preprocessing from that sentence. Both use Amazon Reviews 2014, whereas our study uses AR2023. Hou et al.'s `seq_rec_results/` reference implementation instead uses the `0core_timestamp_w_his_*` HuggingFace subset.

Evaluation practice and reproducibility in recommender systems. Our lead contribution responds to a documented evaluation-trust problem in this field. Ferrari Dacrema et al. [2019] examined 18 recent neural recommenders and could reproduce only 7 with reasonable effort, 6 of which were often outperformed by comparably simple heuristic baselines; the extended journal analysis [Ferrari Dacrema et al. 2021] corroborates the pattern in greater depth and attributes it in part to the choice and optimization of the baselines used for comparison. Sun et al. [2020] respond at the community level with standardized benchmarking for reproducible evaluation and fair comparison. These works diagnose the problem and propose field-level remedies; what this paper adds is complementary and *per-paper*: an executable discipline that a single submission can adopt and a reviewer can re-run — version-controlled pre-declaration of the comparator-facing confirmation (§5.2), a fail-closed artifact gate that recomputes every artifact-gated table cell from released artifacts at build time (published comparator values are pinned as labeled external constants; §2.2 records a retirement the gate forced), local regeneration of the comparator by its own unmodified reference implementation (environment-caveated single-run regenerations, §5.3), and symmetric publication of failures, including a pre-declaration VOIDed by its own comparability check (Appendix A.0) with a redesigned successor that subsequently passed on fresh seeds (`PREREG_OFFICE_V3.md`; §5.2), the original VOID standing unchanged. None of this apparatus strengthens any empirical claim — every non-claim of §5–§7 stands unchanged; it certifies that the numbers printed are the numbers the artifacts recompute, under decision rules frozen before the confirmatory runs.

2.1 Attribution table

Component	Source	Where in our work
SASRec architecture	Kang and McAuley [2018]	base `SASRecSBERT` model
BERT4Rec	Sun et al. [2019]	one of 20 ablation baselines
MiniLM `all-MiniLM-L6-v2` (384-d)	Reimers and Gurevych [2019]; Wang et al. [2020]	default frozen item-text encoder
BLaIR `blair-roberta-base` (768-d)	Hou et al. [2024] (arXiv:2403.03952)	alternative frozen item-text encoder
2-layer MLP adaptor `[768, 300, 64]` Dropout+ReLU	Hou et al. [2024] (SASRecText, in `external/AmazonReviews2023/seq_rec_results/`)	`--mlp-adaptor` flag
Rich-text content concatenation (title + categories + store name)	derived from Hou et al. [2024] preprocessing	`encode_richtext_5core.py`
Amazon Reviews 2023 dataset	Hou et al. [2024]	all benchmarks
TIGER / LIGER	Rajput et al. [2023] / Yang et al. [2024]	published comparator numbers

2.2 Our additions to the above prior art

We do not claim a wholly new recommender architecture. Our architectural additions are deliberately small: a causal FIR adaptation of prior frequency-filter ideas and a soft text-prototype additive term. Our contributions are:

- An **open-source iterative user-and-item 5-core preprocessing pipeline** (`preprocess_5core_standard.py`). Its leave-out chronology resembles TIGER and LIGER and its two-sided filtering resembles LIGER, but it does not claim exact TIGER/LIGER protocol identity and differs from Hou et al.'s 0-core repository path.
- A **memory-efficient chunked-full-softmax loss** that enables proper full-catalog cross-entropy training at 207k-item catalogs on a single 16 GB GPU.
- An **eval optimization** (caching `all_item_features()` once per `evaluate()` call) that gives ~10× speedup at d=64 with the MLP adaptor.
- A **cross-pipeline transfer study**: empirically testing which published architectural choices help on the 5-core protocol.
- **Negative-result documentation** for 18 of 20 tested variants on Beauty_and_Personal_Care 5-core.
- The **competitive Video_Games 5-core result** on the HSTU-style pure-PyTorch stack (6-seed NDCG@10 = **0.0673 ± 0.0003**, +17.5% over published SASRec 0.0573), shown to be architectural (ID-only ≈0.0656), with no SOTA claim. *(The earlier v1-era SASRec-SBERT number 0.05509 ± 0.00035 has been retired from Table 1a: its raw per-seed artifacts predate the artifact manifest and cannot be recomputed, so it survives only as a RETIRED provenance row in the manifest, not as a paper claim.)*

- The two added components — the **strictly causal FIR temporal filter** (positive matched identity contrasts on three outcome-visible categories plus a positive outcome-known Software robustness contrast; learned FIR separates from one equal-parameter compound non-temporal placebo in MI, while the shared causal control prevents per-channel-tap attribution; §5.2) and **TAPE** — plus the **MI frequency-5 tail case** (§5.3) and its **interaction-thinning titration** level-contrast result (§5.3). The novelty boundary for each is stated explicitly in Table 0 (§2.3).

2.3 Claim boundary

Table 0 separates prior work from the paper's actual claim boundary. Frequency filtering (including FEARec, FreqRec, WEARec, and WPGRec) and causal convolution (including Caser and NextItNet) predate this work [Du et al. 2023; He et al. 2026; Liu et al. 2026a; Tang and Wang 2018; Xu et al. 2026; Yuan et al. 2019], as do residual identity initialization, HSTU, frozen text encoders, and label smoothing. Our claim is limited to the gradient-active left-causal depthwise-FIR residual used before an HSTU-style stack under an all-position objective, together with its matched internal evaluation. The historical zero-gated realization is singular at initialization and remains package-attributed; the nonsingular canonical realization permits a FIR-arm interpretation, while the learned FIR separates from one equal-parameter current-position-only DCT/GELU placebo, but that compound contrast does not isolate temporal access; a shared causal filter prevents attribution to per-channel taps (§3, §5.2).

Table 0: Claim boundary — reused basis, specific change, and supported evidence.

Component and reused basis	Specific change	Evidence and boundary
HSTU base — Zhai et al. (2024): pointwise `silu(QK`$^\top$` + rab) V`, per-block normalization, and relative biases	Pure-PyTorch implementation; two spurious normalizations removed	Core-block parity only at a mirrored aligned configuration; no pinned end-to-end reproduction (§3.2, §5.3, §6.3). Prior-work implementation.
BLaIR/SBERT text — Hou et al. (2024), Reimers & Gurevych (2019), Wang et al. (2020)	Frozen off-the-shelf text features and Hou et al.'s MLP adaptor; no method change	Table 1a encoder comparison. Prior work.
AlphaFuse-style text+ID package — Hu et al. (2025)	Official upstream AlphaFuse/SASRec classes under one frozen configuration, with MiniLM-384 substituted for the published text vectors and the paper's complete-history-masked evaluator	AlphaFuse-style NDCG@10 0.048273 [0.048129,0.048416] versus zero-initialized upstream-class SASRec-ID control 0.039024; delta +0.009249 [+0.008329,+0.010169], but -0.019065 versus the existing 0.067337 reference. A separately frozen parser-default Normal(0,1) sensitivity scored 0.043065 [0.042645,0.043486]; the package remains above it by +0.005207 [+0.004779,+0.005635]. Countable outcome-known same-investigator whole-package evidence; not a published-table reproduction, null-space-fusion isolation, equal architecture/capacity/tuning, independent confirmation, or SOTA.

Component and reused basis	Specific change	Evidence and boundary
Label smoothing/time bias — Szegedy et al. (2016); TiSASRec [Li et al. 2020]; HSTU [Zhai et al. 2024]	Integrated with this backbone and full-catalog chunked softmax	+0.0013 and +0.0027 single flags (Table 1). Prior work.
Sequence/frequency and linear-time operators — FMLP-Rec, BSARec, FreqRec, WEARec, TimeWeaver, TV-Rec, HyenaRec, ConvRec, and Mamba4Rec; causal convolutions include Caser, NextItNet, C3SASR, and AdaMCT	The official WEARec model/training code is evaluated under our split, complete-history mask, full-catalog evaluator, cutoff, and tie rule; the others are not inserted as-is	WEARec official-code/equal-evaluation mean 0.059184 [0.058674,0.059693] versus reference 0.067337; `WEAREC-BELOW-EXISTING-REFERENCE`. Outcome-known same-investigator feasibility baseline, not equal architecture/loss/schedule/tuning, independent confirmation, or SOTA. Filtering, frequency modeling, and causal convolution are prior art.
Causal FIR adaptation (ours) — the filtering/convolution line above	Minimal leak-free, gradient-active, identity-initialized left-causal depthwise residual before an HSTU-style all-position stack	Outcome-known FIR—identity estimates: MI +0.002265, IS +0.002110, CDs +0.006150. Software robustness: +0.005062 [+0.004591,+0.005533]. Learned beats the equal-parameter current-only placebo by +0.001941 but does not separate from shared FIR (-0.000081 [-0.000337,+0.000175]). Prospective MovieLens: learned minus identity +0.000000 [-0.000074,+0.000075] and learned minus pointwise +0.000035 [-0.000057,+0.000127]; neither gate passed, so parsimonious-arm NI is conditional only. **Incremental modular contribution; no general FIR benefit or independent confirmation.**
TAPE (ours) — TIGER, VQ-Rec, ProtoMF; prototype/semantic-ID motivation	Frozen soft assignments gate a zero-initialized additive prototype table	+0.0009 single flag; +0.0004 four-seed check. Supporting ablation (§5.1).
Tail/thinning analysis (ours) — standard popularity strata, MELT, DropoutNet [Volkovs et al. 2017], CLCRec [Wei et al. 2021]	Per-dataset text–ID tail contrast and matched-R1 interaction/user thinning	MI +0.000420 (outcome-visible); VG null; cross-dataset p=.13; user-mode p=.058. Secondary empirical boundary (§5.3–§5.3).

Closest filtering and linear-time operator systems. FMLP-Rec and BSARec establish learnable sequence filtering and frequency-rescaled attention; BSARec's Theorem 1 concerns repeatedly applied softmax attention, not our HSTU-style operator. C3SASR and AdaMCT already combine causal/local convolution with attention [Chen et al. 2022; Jiang et al. 2023]. TimeWeaver combines a reparameterized large-kernel convolution with time-aware augmentation and dual temporal/EMA streams [Liu et al. 2025]. TV-Rec replaces fixed kernels and self-attention with position-specific time-variant filters [Shin et al. 2025]; HyenaRec parameterizes long convolution kernels with Legendre polynomials and short-term gates [Liu et al. 2026b]; ConvRec uses hierarchical strided convolution for attribute-aware sequence aggregation [Elsayed et al. 2026]; and Mamba4Rec applies a selective state-space operator with a local convolutional path [Liu et al. 2024a]. Other nearby filtering systems include DLFS-Rec, DIFF, TASIF, and ConvFormer [Kim et al. 2025; Liu et al. 2023; Luo et al. 2026; Wang et al. 2023]; current frequency systems FreqRec and WEARec add inter-session spectral paths or adaptive wavelet filtering [He et al. 2026; Xu et al. 2026]. These systems differ in backbone, attributes, sequence length, candidate protocol, and efficiency target, so their reported numbers are not inserted as if protocol-matched. Their existence rules out broad filtering, convolution, temporal-specificity, or efficiency novelty. Our distinction is only the small gradient-active, identity-initialized **left-causal depthwise FIR residual** placed before one HSTU-style all-position, full-catalog stack, together with the bounded evaluation reported here.

Closest text/ID systems. AlphaFuse [Hu et al. 2025] learns ID embeddings in the null space of language embeddings and is the closest frozen-text-plus-ID comparator. The first AlphaFuse-style MiniLM port on our Video_Games split used mismatched seen-item masking; its repaired V2 factorial remained outcome-visible, capacity/initialization-confounded, incompletely masked for long histories, and non-rank-reconstructive, so its governed adjudicator permanently marks it `NONCOUNTABLE`. Separately frozen V3 used eight fresh seeds per arm, complete-history-masked validation selection, and sealed TEST evaluation. Its verdict, `EEV3-REPORTABLE-OUTCOME-KNOWN`, records AlphaFuse-style MiniLM NDCG@10 0.048273 [0.048129, 0.048416] versus 0.039024 [0.038106, 0.039941] for a zero-initialized upstream-class SASRec-ID control: descriptive independent-arm difference +0.009249 [+0.008329, +0.010169]. The package nevertheless remains -0.019065 [-0.019347, -0.018783] below our existing 0.067337 reference. Separately frozen V4 tested parser-default `Normal(0,1)` SASRec-ID and scored 0.043065 [0.042645, 0.043486]; the package remained above it by +0.005207 [+0.004779, +0.005635], while normal initialization improved SASRec-ID over the earlier zero-init control by +0.004042 [+0.003089, +0.004995]. Official recipes may override initialization by dataset, so the parser setting is not a universal upstream recipe. V4 is outcome-known, same-investigator, and cross-campaign: phase/date and initialization are confounded, and architecture, capacity, and allocation remain unequal. These studies do not reproduce AlphaFuse's published text encoder, isolate null-space fusion or initialization, establish equal architecture/capacity/tuning, provide independent confirmation, or support SOTA. Whitening, channel separation, prototype/codebook, decoupled fusion, and adaptive semantic/ID methods remain active prior lines [He et al. 2025; Hu et al. 2026; Lee et al. 2026; Li et al. 2025; Liu et al. 2021, 2024b; Wei et al. 2026; Xie et al. 2022; Xu et al. 2024; Zhang et al. 2024, 2026; Zhou et al. 2026]. Our text findings remain construction-specific; item-text permutation remains open. Mecos and RecGPT address different few-interaction or zero-shot/short-history settings [Jiang et al. 2025; Zheng et al. 2021].

Two concurrent 2026 preprints reinforce that this is an active modular-content line rather than a priority claim: SISA-Rec injects frozen BERT semantics through gated input fusion and a semantic attention term [Abbasi et al. 2026], while ASER adds review-distilled sensory representations to four existing sequential backbones [Yoon et al. 2026]. Their Amazon Reviews 2014 protocols and representation interventions are not numerically interchangeable with our AR2023 FIR study; we cite them for mechanism-family coverage only.

Tail and reproducibility boundary. SimRec, MELT, DropoutNet, CLCRec, and LLM-ESR already address sparse, long-tail, or content-supported recommendation [Brody and Lagziel 2024; Kim et al. 2023; Liu et al. 2024b; Volkovs et al. 2017; Wei et al. 2021], so our tail contribution is a measured, dataset-specific frequency-5 case rather than a method claim. LLM2Rec uses Amazon Reviews 2023 with 5-core leave-one-out/full ranking, but its length-10 histories and resulting item universe prevent direct numeric comparison [He et al. 2025]. Elliot, DaisyRec 2.0, accountability workflows, and reproduction audits predate our apparatus [Anelli et al. 2021; Bellogín and Said 2021; Petrov and Macdonald 2022; Sun et al. 2022]; we claim only this paper's combination of per-cell recomputation, provenance tripwires, and symmetric self-VOIDing, not evaluation infrastructure generally. Chunked exact cross-entropy is likewise implementation engineering, not a contribution [Wijmans et al. 2025].

3 Method

3.1 Preprocessing pipeline

We preprocess Amazon Reviews 2023 [Hou et al. 2024] into 5-core leave-last-out splits. For each category we:

1. Load raw user-item review interactions from the `raw_review_<category>` HuggingFace subset.
2. Apply a 5-core filter iteratively until no user or item has fewer than 5 interactions.
3. Sort each user's interactions chronologically.
4. Hold out the last interaction as test, the second-to-last as validation, the rest as training.
5. Encode item metadata (`meta_<category>.jsonl`) with the chosen frozen text encoder.

This protocol shares chronological leave-last-out structure with TIGER [Rajput et al. 2023] and LIGER [Yang et al. 2024], but it is not identical to either: our iterative filtering applies to users and items, matching LIGER's stated k-core dimension. The TIGER paper states user filtering, but the exact geometry of its released preprocessing is not established here; both comparator papers use Amazon Reviews 2014 rather than AR2023. Hou et al. [2024]'s AR2023 reference repo (`external/AmazonReviews2023/seq_rec_results/`) uses the `0core_timestamp_w_his_*` HuggingFace subset. These pipelines produce different catalogs and cannot support direct numerical comparisons.

Estimand and construct boundary. Each review event is treated as an implicit positive interaction regardless of its 1–5-star rating or verified-purchase flag. The estimand is therefore *ranking the next recorded review event among the fixed post-filter catalog*, not predicting preference, satisfaction, purchase, or deployment engagement. Five-core eligibility, item indices, and the candidate catalog are computed from complete histories before splitting, so future events can affect inclusion even though their identities are not used as training targets. Rating-threshold, verified-purchase, genuine implicit-event, and global-time/query-time-catalog sensitivities have not been run; conclusions do not extend to those constructs or deployment regimes.

Pre-core deduplication (disclosed 2026-07-19). Before k-core filtering, duplicate `(user, item)` interactions are collapsed to a single event, keeping the **earliest** timestamped occurrence (first-exposure convention). This step is material and was previously undocumented in the paper: it removes 41,888 of 3,017,439 raw rows on Musical_Instruments (1.39%), 69,115 of 4,624,615 on Video_Games (1.49%), 320,078 of 23,911,390 on Beauty_and_PC (1.34%), and 156,363 of 12,845,712 on Office_Products (1.22%); the FIR-breadth categories were preprocessed from AR2023 rating-only exports whose (user, item) pairs are already unique (0 rows removed on Industrial_and_Scientific and CDs_and_Vinyl). Deduplication can change k-core membership, sequence order, and held-out targets; sensitivity to a keep-latest rule is untested (a disclosed limitation). (The MI/Office counts are from the retained preprocessing logs; the VG/Beauty logs were not retained, and their counts were regenerated 2026-07-19 from the raw dumps with the released deterministic script.)

Timestamp-tie policies (disclosed 2026-07-19; two distinct deterministic rules). Two tie-breaking rules operate at different stages and were previously undocumented. (i) Preprocessing stable-sorts each user's interactions by timestamp only, so for equal-timestamp events the raw-file order decides both which duplicate survives deduplication and which event lands on a leave-two-out split boundary. (ii) The trainer later re-sorts each user's train sequence by `(timestamp, item_index)`, so within-train equal-timestamp order is decided by reindexed item ID. Both rules are deterministic given the released raw dumps and code. Measured scale (2026-07-19, from the released splits): on Musical_Instruments, 3 of 57,439 users (0.01%) have test-target timestamp equal to the validation-target timestamp, 3 have validation equal to the last train timestamp, and 12 (user, timestamp) groups cover 24 of 511,836 rows; on Video_Games the counts are 4 and 8 of 94,762 users and 61 groups covering 122 of 814,586 rows (0.01%). At this scale the tie policies cannot materially affect any reported number; we nevertheless document them as part of the exact preprocessing contract.

3.2 Model architecture

Shared item-feature construction (both encoder families). For each item `i`:

- A learned per-item embedding $e_i \in \mathbb{R}^d$ (initialized N(0, 0.02), padding-aware).
- A frozen text embedding $s_i \in \mathbb{R}^k$ from MiniLM (k=384) or BLaIR (k=768).
- A learnable projection $\Phi : \mathbb{R}^k \to \mathbb{R}^d$.

The item feature passed into the encoder is $f_i = e_i + \Phi(s_i)$. Sequences are right-padded with a dedicated pad token and processed with a causal-only attention mask (no key-padding mask — see §3.4). The output logits for position t are $h_t \cdot$ `all_items.T` where `all_items` is the (n_items, d) item-features table.

Headline encoder — HSTU-style (every §5 result). All headline experiments use the **HSTU-style pure-PyTorch pointwise-attention encoder**, described in full in §3.7: pointwise $\mathrm{silu}(QK^\top + \mathrm{rab})V$ aggregation with the published per-block normalization — no softmax, no $1/\sqrt{d}$ scaling — with the core block verified exactly (max abs diff `0.0`) against the reference research implementation at a mirrored, aligned, dropout-free configuration (`HSTU_PARITY_REPORT.md` discloses the alignment; the trained configuration is a strict-superset HSTU-style variant — §6.3). Configuration, stated here and in §4.4: **4 layers, 2 heads, d = 64, dropout 0.5**.

Baseline encoder — SASRec-SBERT (protocol parity and the superseded Appendix A.1 scans only). The SASRec-family baseline follows Kang and McAuley [2018]: a 2-layer pre-norm Transformer encoder (d=64, n_heads=2, FFN dim=4d, GELU activation) followed by a final LayerNorm, using the same item-feature construction. **No headline number uses this encoder**; it appears as the per-category parity/floor baseline and in the superseded Beauty scan (Appendix A.1).

3.3 Choice of projection Φ

Following our cross-pipeline transfer study (Appendix A.1), we compare two designs for Φ:

- **Linear**: $\Phi(s) = Ws$ for a learned weight $W \in \mathbb{R}^{d \times k}$.
- **MLP** (adopted from Hou et al. [2024] SASRecText): $\Phi(s) = W_2 \cdot \mathrm{ReLU}(W_1 \cdot \mathrm{Dropout}(s))$ with hidden dim 300 and Dropout(0.2) between and before the linear layers. **This design is from Hou et al. [2024] and we do not claim novelty for it.**

3.4 Right-padding + causal-only mask

During exploration we identified a numerical failure mode in the standard SASRec implementation pattern: combining left-padding with both a key-padding mask and a pre-norm Transformer encoder produces NaN logits when an entire row of keys is padded (the softmax denominator is zero). The NaN propagates through residual connections and silently inflates eval NDCG to ~0.9955 — a “fake-perfect” result. We adopt right-padding + causal-only mask (no key-padding mask) to avoid this trap.

3.5 Chunked-full-softmax loss

For training with cross-entropy over the full catalog (no sampled negatives), the full logits tensor `(B, L, n_items)` is prohibitive at L=50 and n_items=207k. We implement an incremental-logsumexp loss:

```
def chunked_full_softmax_loss(hidden, targets, model, item_chunk):
    lse = -inf  # running logsumexp over the item dimension
    target_scores = None
    for chunk_start in range(0, n_items, item_chunk):
        chunk_emb = model.item_features(chunk_ids)
        chunk_logits = hidden @ chunk_emb.T
        lse = logsumexp(lse, logsumexp(chunk_logits, axis=-1))
        if any(target in chunk):
            target_scores[in_chunk] = chunk_logits[in_chunk, target]
    return -mean(target_scores - lse)
```

This bounds peak memory by `(batch, item_chunk)` instead of `(batch, n_items)` and allows proper full-softmax training at the 207k-item Beauty_and_Personal_Care scale on a 16 GB GPU.

Memory-efficient exact cross-entropy has published precedent (Cut Cross-Entropy, ICLR 2025, computes exact CE without materializing the full logit matrix); our chunked variant is independent implementation engineering in the same spirit, claimed only as engineering.

3.6 Evaluation protocol

We score each user against the **full catalog** of 207k items (Beauty_and_PC) or 25k items (Video_Games). For the held-out test item, we compute the rank under dot-product scoring (after masking train+val items the user has already seen) and report NDCG@10, HR@10, MRR. We use the standard SASRec test-time convention of including the validation item in the input sequence [Kang and McAuley 2018].

For our default 16 GB GPU, the optimized evaluation caches the (n_items, d) item-features table once per `evaluate()` call rather than recomputing it per batch — this is critical with the MLP adaptor, which would otherwise recompute the 207k-item × 768 → 300 → 64 MLP forward 1,400+ times per evaluation.

3.7 Added components (deliberately small adaptations)

Our two architectural additions (the novelty boundary is stated in §2.3) are deliberately simple, default-OFF, and zero-initialized so that each is a *bit-identical no-op at initialization* and reduces the design to a single controlled lever. Both are defined on top of an **HSTU-style pure-PyTorch implementation of the HSTU encoder, based on the published architecture** [Zhai et al. 2024]: we use the pointwise-aggregated interaction $\mathrm{silu}(QK^{\top} + \mathrm{rab})\,V$ with the published per-block normalization, having diagnosed and removed two spurious normalizations (a $1/\sqrt{d}$ score scaling and a $1/(i+1)$ row averaging) in our initial reimplementation that collapse HSTU's pointwise attention into weak mean-pooling. We verify the implementation equation-by-equation against the published HSTU formulation (pointwise $\mathrm{silu}(QK^{\top} + \mathrm{rab})\,V$ aggregation, no softmax, no $1/\sqrt{d}$ scaling); an executable end-to-end reproduction of the official kernel stack is impossible on our hardware (§5.3); the core block, however, passes an exact numerical parity test against the reference research implementation at a mirrored **aligned configuration only** (identity affine norms, zeroed extra `uvqk` bias, ε aligned, dropout off, eval mode; max abs diff 0.0; `HSTU_PARITY_REPORT.md` discloses the alignment — the trained configuration is a strict-superset HSTU-style variant, §6.3). **Initialization disclosure (2026-07-19):** with the delta-kernel and gate g=0 initialization, both task gradients are exactly zero at the start (y=x, so the gate gradient vanishes and the kernel gradient is multiplied by g=0); learning bootstraps only because Adam's coupled weight-decay term (1e-5) perturbs the delta tap off its fixed point — the optimizer regularizer is therefore part of the filter's learning mechanism, and with weight decay disabled this parameterization is an absorbing no-op. The fixed-average and no-gate comparator arms do not share this singular start, so those comparisons alter the optimization path as well as the named design element. The later nonsingular studies provide cleaner internal contrasts but reuse outcome-visible splits, and active controls prevent uniquely learned-tap attribution; they do not retroactively decompose the non-initialization-matched historical arms.

(a) Text-Anchored Prototype Embeddings (TAPE). TAPE is a simple soft text-prototype additive capacity term inspired by semantic-ID and prototype methods (TIGER, VQ-Rec, ProtoMF); it provides a small but not headline gain (§5.1). Let $t_i \in \mathbb{R}^{d_{\text{text}}}$ be item i's frozen text embedding (MiniLM or BLaIR). We k-means the L2-normalized text embeddings into K centroids $\{c_1, \dots, c_K\}$ once, offline, and form a **frozen soft assignment** $A_i = \text{softmax}(\langle t_i, c \rangle / \tau) \in \Delta^K$ (τ = 0.05). We add a **learnable prototype table** $P \in \mathbb{R}^{K \times d}$ (zero-initialized) to each item's feature:

$$e_i = \text{id}_i + \Phi(t_i) + A_i P,$$

where id_i is the learned per-item embedding and Φ the learned projection of the frozen text embeddings. Items in the same semantic neighborhood share the learnable capacity $A_i P$ — the shared-structure benefit of TIGER-style semantic IDs [Rajput et al. 2023] without discrete codes or an autoregressive decoder. Because A is frozen and P is zero-init, TAPE is an exact no-op at initialization and recovers the baseline. TAPE is adjacent to TIGER (discrete RQ-VAE IDs), VQ-Rec [Hou et al. 2023] (PQ codes), and ProtoMF [Melchiorre et al. 2022] (prototype matrix factorization for explainability); it differs only in realization — a frozen text-derived soft assignment gating a learnable prototype table inside a sequential recommender — and we present it as a small adaptation of these prior ideas tested as a secondary ablation, not as a headline contribution. Default K = 512.

(b) Canonical causal FIR temporal filter (the proposed module). BSARec [Shin et al. 2024, Theorem 1] proves a low-pass limit for repeatedly applied softmax self-attention. That result does not cover the non-softmax HSTU-style pointwise operator used here and supplies motivation only. We test whether an explicit local left-causal residual benefits this backbone. Learned FIR separates from one equal-parameter current-position-only DCT/GELU residual, a compound contrast that does not isolate temporal access, while a competitive shared causal filter prevents attribution to per-channel taps (§5.1–§5.2). The proposed module is a single **per-channel learnable FIR residual** on the sequence embeddings $x \in \mathbb{R}^{B \times L \times d}$, in a **nonsingular, gradient-active** parameterization:

$$x \leftarrow x + \text{DWConv}_\Delta(\text{pad\_left}(x, K-1)), \qquad \Delta = 0 \text{ at init},$$

implemented as a depthwise `Conv1d(d, d, kernel_size=K, groups=d, bias=False)` whose tap residual Δ is zero at initialization, with a **left-only causal pad of** $K-1$, so the layer is an **exact identity at initialization** yet has **nonzero gradients on the first task update** — there is no gate whose zero initialization creates an absorbing fixed point. The primary E-A and canonical-breadth learned arms used the backbone weight-decay setting for FIR taps; E-A's zero-FIR-weight-decay sensitivity was not detectably different (A2–A1 +0.000010 [-0.000339, +0.000360], p = .95), which is neither an equivalence result nor a pathway/mediation test (§5.2). The left-only pad makes output position t a function of input positions $\leq t$ only — so under our all-position next-item loss there is **no future-label leakage**. FMLP-Rec [Zhou et al. 2022] and BSARec use bidirectional sequence filters in their original formulations; under our all-position next-item loss, directly inserting such filters before each position's prediction would mix future positions, so we use this left-causal realization. **Legacy zero-gated realization (historical runs; reported as package-level).** The Video_Games headline configuration (§5.1) and the original breadth campaign (§5.2) were trained with a functionally related but *singular* realization — $x \leftarrow x+g\odot(y-x)$ with a delta kernel and a scalar gate $g = 0$ at initialization — in which both task gradients vanish at the start, so learning could begin only after Adam's weight-decay term perturbed the delta tap (§3 initialization disclosure). We call that the **legacy zero-gated FIR package** and report its contrasts as package-level (FIR-plus-initialization/optimizer), not FIR-specific. The nonsingular canonical form was then evaluated under matched per-seed initialization on three already outcome-visible category/split settings. Those internal estimates are positive, but their test exposure prevents an independent-confirmation claim. On Musical_Instruments, a later equal-parameter pointwise placebo discriminates learned FIR from that compound current-only residual without isolating temporal access, and the shared causal control prevents learned-per-channel-tap attribution (§5.2). The novelty boundary is narrow and we state it plainly. *Not new*: filtering sequential representations, causal convolution (Caser; NextItNet), the frequency-domain motivation, and the attention-oversmoothing framing (FMLP-Rec; BSARec). *New*: the minimal, gradient-active, identity-initialized, leak-free left-causal depthwise-FIR residual inserted before an HSTU-style stack under the all-position next-item objective, evaluated under full-catalog LLOO. As an FIR filter the module has a learnable per-channel frequency response — the only sense in which we use the word "spectral" — and supplies an explicit local temporal path; we do not claim the attention stack cannot represent this behavior. The legacy-package gain is robust across kernel lengths K ∈ {4, 8, 16, 50} (§5); the canonical primary configuration fixes **K = 16** (historical runs used K = 8 on Video_Games and K = 16 on Musical_Instruments, §5.1–§5.2).

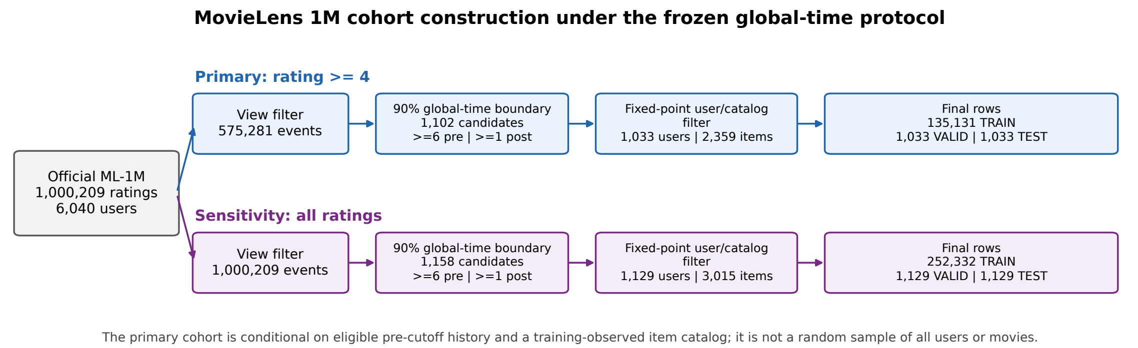
A separate, subsequently frozen Software attempt also evaluated the canonical K=16 form and returned a positive thresholded verdict. Because earlier V2 validation-output non-visibility is not independently established, it is classified outcome-known/exploratory same-team robustness; its same-investigator, same-code-lineage, same-Amazon-family, local-custody scope also precludes independent-confirmation wording (§5.2).

Both components are leak-free by construction and add negligible parameters (TAPE: K×d ≈ 33k; filter: K×d ≈ 0.5–3k). Two further regularizers we use are *not* claimed novel: **label smoothing** [Szegedy et al. 2016] on the full-softmax CE target, and the standard time / text-similarity / relative-position attention biases (TiSASRec [Li et al. 2020]; HSTU [Zhai et al. 2024]; Shaw et al. 2018), each cited at the point of use.

4 Experiments

4.1 Datasets

The primary development and comparator studies use AR2023 5-core categories under the full-catalog LLOO protocol (§3.6). A separate prospective robustness and efficiency study uses MovieLens 1M [Harper and Konstan 2015] under a frozen global-time split. AR2023 counts are total 5-core interactions; the MovieLens row reports its filtered primary view and training-observed catalog:



MovieLens cohort flow (unnumbered). The primary estimand is conditional on eligible pre-cutoff history and a training-observed item catalog; it is not a random sample of all users or movies.

Category	Role in this paper	Users	Items	Interactions (total)
Video_Games	descriptive system context (§5.1); tail-pattern null (equivalence claim retracted §5.3)	94,762	25,612	814,586
Musical_Instruments	outcome-known FIR internal study + pre-declared per-category point-estimate comparison (§5.2)	57,439	24,587	511,836
Office_Products	V1 prereg **VOID** (descriptive; Appendix A.0); **redesigned V3 prereg PASSED** (§5.2; `OFFICE_V3_RESULTS.md`)	223,308	77,551	1,800,878
Beauty_and_Personal_Care	tail-pattern exploratory 3-seed estimate (§5.3); superseded cross-pipeline scan (Supplement S.1)	729,576	207,649	6,624,441
Industrial_and_Scientific	**pre-declared FIR-breadth: artifact-PASS** (frozen rule fired; paired premise withdrawn; post-hoc independent-arm estimate — §5.2; `FIR_BREADTH_RESULTS.md`)	50,985	25,848	412,947
CDs_and_Vinyl	**pre-declared FIR-breadth: artifact-PASS** (frozen rule fired; paired premise withdrawn; post-hoc independent-arm estimate — §5.2; `FIR_BREADTH_RESULTS.md`)	123,876	89,370	1,552,764
MovieLens 1M, rating≥4	**prospectively frozen non-Amazon FIR replication/efficiency study; negative verdict** (§5.2; Harper & Konstan, 2015)	1,033	2,359 training-observed	575,281 filtered events; 135,131 train rows

Video_Games/Musical_Instruments/Office_Products match the HSTU-BLaIR comparator pipeline's own statistics exactly on users/items (interactions within ± 1; §5.2). The Beauty_and_PC catalog is 8× larger than Video_Games, with median user history of only 5 interactions — a harder benchmark. Every pre-declared campaign that has completed is mechanically adjudicated (verdicts at the relevant result blocks: §5.2, §5.3, §5.3, and Appendix A.0).

The MovieLens primary cohort is explicitly conditional rather than population- representative: after the rating and global-time filters, 1,102 candidate users enter the fixed-point training-catalog filter and 1,033 remain. The all-ratings branch is a construct sensitivity, not a second independent dataset.

4.2 Baselines

- Popularity (trivial floor)
- SASRec with sampled-512 softmax (our "original" baseline)
- SASRec with chunked-full-softmax (improved baseline)

- BERT4Rec [Sun et al. 2019]
- Published comparators: TIGER [Rajput et al. 2023], BLaIR [Hou et al. 2024], LIGER [Yang et al. 2024]

Comparator-design matrix. This matrix separates the dimensions that are actually aligned from those that remain unmatched; a shared evaluator is not treated as an equal-model experiment.

Comparator use	Aligned dimensions	Known unmatched dimensions	Supported scope
FIR vs identity, Amazon canonical campaigns	same data, evaluator, backbone, schedule, and exact per-seed backbone initialization	+1,024 FIR parameters; outcome-known categories; TEST access differs by campaign	internal modular contrast on the named settings
Learned FIR vs pointwise placebo, MI	same data/evaluator/backbone/init and exactly 1,024 trainable component parameters	temporal history versus a compound DCT/GELU/current-position operator; outcome-known	learned FIR versus this tested placebo, not temporal access alone
Learned FIR vs shared causal filter, MI	same data/evaluator/backbone/init and causal-history access	1,024 per-channel versus 16 shared parameters; outcome-known	per-channel-tap necessity not established
MovieLens six-arm study	same frozen split/evaluator/backbone/init, schedule, seed blocks, and sealed endpoint timing	parameter counts and filter parameterizations differ; same investigator; record-level endpoints private	negative transfer test; parsimonious-arm NI is conditional only
Official WEARec under our evaluator	official model/training code; same split, mask, catalog, cutoff, and tie rule	architecture, loss, schedule, capacity, and tuning budget unequal; private endpoints	equal-evaluation feasibility baseline only
AlphaFuse-style package	official upstream classes; one frozen configuration; same split/evaluator	MiniLM substitutes for published vectors; architecture, capacity, initialization, and tuning history unequal	whole-package comparator coverage only
Published/local HSTU-BLaIR reference	users/items match and interactions differ by one; local regeneration uses the reference code	single-run/unpaired reference; pinned environment unavailable; no equal tuning budget	per-category point threshold and environment-caveated context

4.3 Experimental design (headline runs)

All headline results (§5.1–§5.3) use the HSTU-style pure-PyTorch encoder (§3.2) under the fixed AR2023 5-core full-catalog LLOO protocol (§3.6). Each configuration is trained for **40 epochs** (the §5.1 Video_Games headline; the pre-declared-file MI/Office V3/FIR-breadth campaigns train 20 epochs per their frozen configurations) at batch size 256 (d_model 64, 4 layers, 2 heads, dropout 0.5) with the memory-efficient chunked-full-softmax loss (§3.5, item chunk 32,768) and a warmup-cosine learning-rate schedule. On top of the bias stack (TAPE-512 + time bias + text-similarity bias + pos-rab) the winning configuration adds label smoothing (ε=0.2) and the causal FIR filter (K=8). We run **6 seeds (20260608–20260613)** for the full-model headline and **5 seeds (20260608–20260612)** for every per-component ablation rung, reporting mean ± sample-std; the reported test number is always selected by best validation NDCG@10 (best-by-val), never by test. **Fixed-split inference scope (stated prominently, 2026-07-20):** the training seed is the only randomized unit; every interval in this paper quantifies optimization variability on one fixed public split per category — nothing here estimates user-resampling, split-choice, category-sampling, or cross-dataset uncertainty, and all claims are scoped to these exact datasets and splits. Evaluation is full-catalog (n_eval = 94,762 on Video_Games), with all train+val items masked (§3.6).

The tail analysis (§5.3–§5.3) rests on two arm comparisons, each toggling one named configuration flag on the same split and seed numbers (bundled interventions: the arms are not initialization-paired and optimizer paths differ) and reporting `by_popularity` NDCG@10 / HR@10 on leak-free train-frequency terciles (terciles frozen at full density):

- **text vs ID-only:** the full text stack against an ID-only ablation (`--no-sbert`, no prototypes, no text-similarity bias).

- **density / connectivity titration (§5.3):** training-only sub-sampling that thins either interactions (interaction-mode) or whole users (user-mode) to match a sparser reference category's resource levels, with the evaluation set held fixed.

A separate **20-variant cross-pipeline transfer scan on Beauty_and_PC** — supporting reproducibility material, not part of the headline spine — is reported in Appendix A.1.

The MovieLens study used six exact matched-initialization arms over eight pre-specified seed blocks: identity, channel-shared K=16, eight-group K=16, rank-at-most-8 tangent-factorized K=16, per-channel K=16, and an equal-parameter current-position-only placebo. Acquisition had already read the official ratings file, constructed and hash-bound the TEST split, and used target-in-training-catalog eligibility to define the primary cohort. During fitting and validation selection, trainers consumed TRAIN+VALID only and emitted no TEST scores. After all 96 training runs completed, each selected checkpoint received one sealed TEST evaluation. The primary estimand was NDCG@10 on rating-at-least-4 events after a global 90% time boundary; all ratings, user/item cluster bootstraps, and resource readings were frozen sensitivities. The three-candidate noninferiority family used margin 0.000500 and Holm-adjusted one-sided tests.

4.4 Training details and hardware

Headline runs use Adam (lr 1e-3, weight decay 1e-5) with gradient clipping (norm 5.0) under the 40-epoch warmup-cosine schedule above. Hardware: a single NVIDIA RTX 5060 Ti (16 GB, Blackwell sm_120). Video_Games training takes ~10 min/seed; Beauty_and_PC ~1.5–2 hr/seed with the chunked-full-softmax loss. (The Appendix A.1 Beauty scan used a shorter 15–20-epoch budget and seeds 42/43; those details are local to that supporting study.)

5 Results

AlphaFuse pipeline boundary. Because the upstream offline preprocessing code was unavailable, both official classes consumed training rows produced by our frozen per-prefix adapter. This is upstream-class/training-code transfer, not reproduction of the upstream data pipeline.

Statistical reporting conventions. Unless stated otherwise: the sample unit is the training seed; means are reported ± sample standard deviation over seeds; confidence intervals are two-sided 95% Student-t intervals with df = n−1; comparisons between our own arms use same-numbered seeds and are analyzed as independent arms (Welch/Satterthwaite) unless exact matched backbone initialization and a paired analysis are explicitly named, as in canonical breadth, active controls, pointwise placebo, and prospective Software V3; comparisons against published numbers are point-estimate comparisons; pooled tail hit-counts are descriptive counts only; p-values are uncorrected unless a correction is named. Cells labeled *confirmatory* in the artifact manifest are pre-declared prospective campaigns or frozen-protocol locks: selection timing and protocol integrity, not seed count or external independence, are the criterion. Multi-seed post-hoc results remain exploratory.

5.1 Video_Games — descriptive system context (not FIR inference; NOT SOTA)

The complete historical component ladder and retired-baseline context are retained in the canonical reader/reviewer supplement. The main article keeps only the current comparator evidence needed to locate the modular FIR claim:

the published HSTU-BLaIR anchor, the official-code WEARec feasibility result, and the frozen AlphaFuse-style whole-package comparison. None is an equal-model experiment or the inferential basis for the matched FIR contrasts in §5.2.

Table 1b: Reported, locally regenerated, and current-comparator evidence [Hu et al. 2025; Liu 2025; Xu et al. 2026]:

The HSTU-BLaIR paper reports the same Video_Games item and user counts as ours (25,612 items / 94,762 users), but 814,585 interactions—one fewer than our 814,586—under its AR2023 5-core LLOO geometry. They train for **100 epochs** following Zhai et al.'s HSTU protocol. Their single-seed numbers:

Method	NDCG@10	HR@10	Comparison to ours
SASRec [Liu 2025, single seed, 100 epochs]	**0.0573**	0.1028	−15% below our full model 0.0673 ± 0.0003
HSTU [Zhai et al. 2024]	0.0741	0.1315	+10% above our full model
HSTU-OpenAI (TE3L)	0.0742	0.1328	+10% above our full model
HSTU-BLaIR [Liu 2025]	**0.0760**	**0.1353**	**+13% above our full model; stronger reported reference**
HSTU-BLaIR local SM120 compatibility port (this audit)	0.07382 final full-eval (exported artifact; the higher best-epoch reading survives only in an unretained WSL log and is excluded from the artifact graph)	0.13234 final	stronger than ours; not a faithful pinned-environment reproduction
AlphaFuse-style MiniLM representation package (8 fresh seeds; this audit)	0.04827 [0.04813, 0.04842]	0.08976 [0.08935, 0.09016]	`EEV3-REPORTABLE-OUTCOME-KNOWN`; +0.00925 [0.00833, 0.01017] vs zero-initialized upstream-class SASRec-ID, but −0.01906 [−0.01935, −0.01878] vs our reference
Zero-initialized upstream-class SASRec-ID control (8 fresh seeds; this audit)	0.03902 [0.03811, 0.03994]	0.07363 [0.07235, 0.07491]	frozen V3 ID-only control; parser-default `Normal(0,1)` not tested; official recipes may override initialization by dataset; not paired with the package arm
Parser-default `Normal(0,1)` upstream-class SASRec-ID sensitivity (8 fresh seeds; this audit)	0.04307 [0.04265, 0.04349]	0.07931 [0.07871, 0.07990]	`EEV4-ALPHAFUSE-ABOVE-NORMAL-SASREC`; package minus this control +0.00521 [0.00478, 0.00563]; cross-campaign outcome-known sensitivity, not paired or initialization-isolating
WEARec official model/training code (8 assessment seeds; this audit)	0.05918 [0.05867, 0.05969]	—	`WEAREC-BELOW-EXISTING-REFERENCE`; −0.00815 [−0.00869, −0.00762] vs our six-seed 0.06734 reference under the shared evaluator

Our full model (0.0673) exceeds their published SASRec (0.0573) but does not reach their HSTU (0.0741) or HSTU-BLaIR (0.0760) on Video_Games. Our local compatibility-port HSTU-BLaIR run reached final full-eval NDCG@10 `0.07382` (the best-epoch reading exists only in an unretained WSL log and is excluded from the artifact graph); the run is stronger than SASRec-SBERT but carries SM120/fbgemm validity caveats.

The separately frozen WEARec campaign used the official AAAI 2026 repository at commit `2087335339b1` (the full hash is artifact-bound). Two official-domain presets were selected by one tuning seed using validation NDCG@10 only; the selected `official_beauty` preset then trained on eight fresh assessment seeds with TEST scoring suppressed until all checkpoints existed. The committed adjudicator's first authorized endpoint read returned **`WEAREC-BELOW-EXISTING-REFERENCE`**: WEARec mean NDCG@10 **0.059184, 95% CI [0.058674, 0.059693]**, versus the existing six-seed full-model reference **0.067337 [0.067063, 0.067611]**. The descriptive outcome-known unpaired Welch contrast is **-0.008154 [-0.008689, -0.007618],** $p = 1.16 \times 10^{-11}$. This is narrow official-model/equal-evaluation evidence by the same investigators on an outcome-known split—not independent confirmation, SOTA, a paired experiment, or equality of architecture, loss, schedule, or tuning budgets. The graph replays the released NDCG vectors and Welch arithmetic; private endpoint extraction and HR/MRR raw-vector arithmetic are not publicly replayed.

The clean E-E V3 campaign froze the AlphaFuse upstream at commit `b501a0540b60` (the full hash is artifact-bound), the official repository `AlphaFuse` and `SASRec` classes, one shared training configuration, and eight fresh seeds per arm. MiniLM-384 title vectors replace AlphaFuse's published OpenAI text vectors, so this is an AlphaFuse-style port rather than a published-table reproduction. Prelaunch preparation opened the pre-existing outcome-known combined TRAIN/VALID/TEST export but wrote a training input containing only user IDs, TRAIN histories, and VALID targets. Model fitting and complete-history-masked VALID selection did not load, hash, or score TEST; after all 16 selected checkpoints were bound into the family-wide READY record, 16 sealed one-shot TEST evaluations preceded the unchanged committed adjudicator. Its exact verdict, **`EEV3-REPORTABLE-OUTCOME-KNOWN`**, is countable as current-comparator evidence: AlphaFuse-style NDCG@10 **0.048273 [0.048129, 0.048416]**, zero-initialized upstream-class SASRec-ID control **0.039024 [0.038106, 0.039941]**, and descriptive independent-arm Welch difference **+0.009249 [+0.008329, +0.010169],** $p = 3.48 \times 10^{-8}$. Fixed-split user and target-item-cluster bootstrap sensitivities are also positive, but are not optimizer- or population-level inference. The package remains **-0.019065 [-0.019347, -0.018783]** below the existing 0.067337 full-model reference. Because text availability, initialization, trainable capacity, parameter allocation, and architectures differ, and the parser-default `Normal(0,1)` setting was not tested, this is a whole representation-package contrast. Official recipes may override initialization by dataset, so the parser setting is not a universal upstream default. The contrast is not null-space-fusion isolation, equal-tuning evidence, a paired experiment, independent confirmation, or SOTA. The public graph recomputes released aggregate vectors and Welch arithmetic and checks endpoint/sidecar ledger shape, 64-hex syntax, and uniqueness; it does not read or hash the private endpoint/sidecar files or replay record-level bootstraps. The local adjudicator, which had access to the private files, checked their recorded digests.

The separately frozen E-E V4 campaign addressed only that initialization sensitivity. Eight fresh parser-default `Normal(0,1)` SASRec-ID runs completed before eight sealed one-shot TEST evaluations and the committed adjudicator's first endpoint read. Its exact verdict is **`EEV4-ALPHAFUSE-ABOVE-NORMAL-SASREC`**: normal-init SASRec-ID NDCG@10 **0.043065 [0.042645, 0.043486]**, V3 AlphaFuse-style package minus V4 normal-init SASRec-ID **+0.005207 [+0.004779, +0.005635],** $p = 1.02 \times 10^{-9}$, and V4 normal-init minus V3 zero-init SASRec-ID **+0.004042 [+0.003089, +0.004995]**. The normal-init arm remains **-0.024272 [-0.024728, -0.023816]** below the existing reference. This is countable only as prospectively frozen, outcome-known, same-investigator cross-campaign comparator-fairness sensitivity. Phase/date and initialization are confounded; architecture, capacity, and parameter allocation remain unequal. It does not isolate initialization, supply an equal-budget factorial or independent confirmation, or support SOTA. The public graph recomputes the released summary and three Welch contrasts and validates ledger structure without reading private endpoint/sidecar files.

Relative to HSTU-BLaIR, this paper contributes multi-seed internal ablations, a released preprocessing path, and implementation diagnostics—not a leaderboard result. The reported 0.0760 remains the stronger Video_Games reference. Core-block equality at one aligned configuration (§3.2) and environment-caveated local regenerations (§5.3) do not constitute a pinned end-to-end reproduction.

5.2 Canonical causal FIR internal contrasts and untuned category reuse (legacy-package attribution caveat §3)

The strongest second-category package result appears on Musical_Instruments, an already outcome-visible AR2023 category; it is internal robustness evidence, not transfer confirmation. The full V2 stack reaches **NDCG@10 = 0.0413 ± 0.0005 (5-seed)**, **+7.9%** over a matched HSTU+TAPE base (0.0383 ± 0.0004) and **+57%** over a plain ID-only SASRec (0.0264). A two-arm comparison (not a controlled isolation — §3 initialization caveat) (best-by-val, vs the MI SBERT+TAPE base 0.0383, four 20-epoch seeds 20260609–12) separates the two supported regularizers at the arm level (bundled arms; no component isolation):

Table 1c: Musical_Instruments arm-by-arm comparison (NDCG@10, best-by-val, vs MI SBERT+TAPE base; shares are of the 5-seed V2 combined lift +0.0032).

Configuration	NDCG@10	Δ vs base	share of combined lift
MI SBERT+TAPE base (four 20-epoch seeds)	0.0383	—	—
+ label smoothing only (5-seed)	0.0391 ± 0.0001	+0.0008	26%
+ FIR package arm (filter component; attribution open) (k16, 5-seed)	**0.0408**	**+0.0025**	**77%**
+ both = V2 (k16, 5-seed)	0.0415	+0.0032	100%

(Recomputed best-by-val from the released per-seed artifacts. Against the headline 5-seed V2 stack the package arm's share is 77% (k16) / 82% (k8). LS-only is the 5-seed family `results_MI_lsonly_seed{08-12}` (0.03914 ± 0.00013); base is the four e20 seeds 20260609–12.)

The package arm contributes ≈3× what label smoothing does on the second category, and ≈80% of the combined lift in that same-seed decomposition — a package-level share, no component isolation — (77% vs the k16 stack, 82% vs the k8 headline stack) — i.e. the historical package's gain appears on both development categories while label smoothing's does not (no formal component-by-category interaction was tested). Its kernel-length insensitivity also appears on MI (MI k8 0.0413 ± 0.0005 ≈ k16 0.0415 ± 0.0002). Component attribution for these historical runs remains open; the matched-initialization studies below provide cleaner internal contrasts, subject to their outcome-visibility limits.

Pre-declared matched-arm FIR study (`PREREG_FIR_V3.md` / E-A; mechanically adjudicated verdict W-POS; outcome-visible split). A nonsingular reparameterization, $y = x + \mathrm{conv}_\Delta(x)$ with $\Delta = 0$, made the treatment identity-initialized and gradient-active; three arms by eight seed blocks shared one verified initialization within each block. The frozen registered analysis is independent-arm Welch/Satterthwaite: learned FIR minus identity is **+0.002265, ordinary 95% Welch CI [+0.001928, +0.002602]** ($df = 13.939$) on Musical_Instruments. The paired-by-seed difference is retained only as descriptive. This is a cleaner within-configuration contrast than the historical package arms, but the category and split had already informed development and the execution record contains the provenance deviations documented in the audit; it is therefore exploratory internal evidence, not fresh confirmation. The zero-FIR-weight-decay sensitivity is A2–A1 **+0.000010 [-0.000339, +0.000360],** $p = .95$: no difference was detected, but neither equivalence nor exclusion of a weight-decay pathway is established. Artifacts: `results_MI_FIRV3_{arm}_seed{seed}.json` (24 runs) and `fir_v3_adjudication.json`.

Pre-declared canonical reparameterization breadth (`PREREG_FIR_CANONICAL_BREADTH.md`; verdict `CANON-BREADTH-POS`; outcome-known/test-exposed). With one frozen K=16, 20-epoch configuration, zero category tuning, and exact per-seed matched initialization, learned minus identity is **+0.002110, ordinary paired 95% CI [+0.001820, +0.002399]** on Industrial_and_Scientific (paired $t = 17.23$; Holm-adjusted $p = 5.44 \times 10^{-7}$; 8/8 positive) and **+0.006150 [+0.005849, +0.006450]** on CDs_and_Vinyl ($t = 48.39$; Holm-adjusted $p = 8.42 \times 10^{-10}$; 8/8 positive). Supportive registered endpoints point in the same direction: HR@10/MRR differences are +0.003474/+0.001819 on Industrial_and_Scientific and +0.010728/+0.005033 on CDs_and_Vinyl; the broader registered cutoff family was not emitted by the trainer and is not claimed. The categories were selected after favorable legacy-package outcomes and TEST was evaluated each epoch, so these estimates are **descriptive robustness of the canonical parameterization, not independent confirmation or population transfer**. The intervals are ordinary paired-t intervals; Holm adjusts decisions/p-values, not confidence limits. All 32 source JSONs and the adjudication record are now hash-bound in the public artifact inventory.

Pre-declared six-arm active-control study (`PREREG_FIR_CONTROLS.md`; eight matched seed blocks; sealed one-shot final TEST; frozen verdict `CTRL-ACTIVE-CONTROL-SUPPORTED`). The frozen analysis has two separate Holm families; all intervals are ordinary paired 95% intervals. In family A, learned is **+0.002116 [+0.001910, +0.002322],** $p_{\text{Holm}} = 2.04 \times 10^{-7}$; fixed MA **+0.000712 [+0.000509, +0.000915],** $p_{\text{Holm}} = 1.30 \times 10^{-4}$; fixed HP **+0.000708 [+0.000510, +0.000907],** $p_{\text{Holm}} = 1.30 \times 10^{-4}$; channel-shared **+0.002197 [+0.002008, +0.002386],** $p_{\text{Holm}} = 1.08 \times 10^{-7}$; and parameter-matched nonlinear causal **+0.001912 [+0.001657, +0.002166],** $p_{\text{Holm}} = 1.33 \times 10^{-6}$. In family B, learned exceeds fixed MA by **+0.001404 [+0.001085, +0.001723],** $p_{\text{Holm}} = 6.37 \times 10^{-5}$ and fixed HP by **+0.001407 [+0.001089, +0.001726],** $p_{\text{Holm}} = 6.37 \times 10^{-5}$, but the fixed arms are algebraically redundant. A learned-specific advantage is not established over shared (**-0.000081 [-0.000337, +0.000175],** $p_{\text{Holm}} = .477$) or nonlinear (**+0.000204 [-0.000038, +0.000446],** $p_{\text{Holm}} = .174$); retained tests are not equivalence. This study alone lacked a parameter-matched non-temporal placebo. It was designed after the learned–identity outcome was known, ran from a dirty evolving tree, and its sidecars were not independently custodied because of a filename-suffix ignore error; we classify it as outcome-known exploratory mechanism evidence despite its frozen analysis and sealed evaluator. Human pre-adjudication visibility remains an author-verification item.

Pre-declared parameter-matched non-temporal placebo (`PREREG_FIR_POINTWISE_V1.md`; eight matched seed blocks; sealed one-shot final TEST; frozen verdict `POINTWISE-FIR-DISCRIMINATED`). The current-position-only placebo uses a deterministic width-16 orthonormal DCT projection, GELU, and zero-initialized learned 16-to-64 map. It is identity-initialized, gradient-active, and has exactly 1,024 trainable parameters, matching the K=16 depthwise FIR, but cannot access earlier positions. In one frozen three-test Holm family, learned FIR minus identity is **+0.001872 [+0.001737, +0.002007],** $p_{\text{Holm}} = 1.87 \times 10^{-8}$; pointwise minus identity is **-0.000069 [-0.000200, +0.000061],** $p_{\text{Holm}} = .249$; and learned FIR minus pointwise is **+0.001941 [+0.001788, +0.002095],** $p_{\text{Holm}} = 2.40 \times 10^{-8}$. All intervals are ordinary paired 95% intervals; the retained pointwise–identity test is not equivalence. This discriminates the learned temporal FIR from the tested equal-parameter non-temporal residual under matched initialization. It does not establish per-channel-tap necessity because the shared causal filter remains competitive, and it is outcome-known Musical_Instruments mechanism evidence, not independent confirmation or cross-domain replication.

Frozen Software robustness attempt (`PREREG_FIR_PROSPECTIVE_SW_V3.md`; eight matched seed blocks; sealed one-shot final TEST; verdict `SW-V3-PRACTICAL-POS`). The learned and identity arms used exact per-seed matched backbone initialization. Training loaded TEST rows only for the explicitly pre-declared transductive item catalog and emitted no TEST scores; checkpoints were selected solely by validation NDCG@10. After all 16 checkpoints existed, the committed driver created exclusive-created, hash-linked local READY/evaluation seals, evaluated each selected checkpoint once on TEST, and invoked the protocol-designated adjudicator. Mean TEST NDCG@10 was **0.120200 learned versus 0.115138 identity**; learned minus identity was **+0.005062, ordinary paired 95% CI [+0.004591, +0.005533]**, paired $t(7) = 25.39$, two-sided $p = 3.75 \times 10^{-8}$, with 8/8 differences positive. The CI lower bound cleared the frozen +0.000500 reporting threshold, yielding the exact mechanical verdict `SW-V3-PRACTICAL-POS`. Tracked evidence cannot establish whether earlier V2 validation output was observed before V3 was frozen, and local logs cannot prove first human/tool access; absent an author custody statement, we therefore classify V3 as **outcome-known exploratory same-team robustness**, not confirmation. It is also same-investigator, same-code-lineage, and same-Amazon-family work under local same-user custody with no external escrow. The frozen tag additionally binds a CRLF raw digest for one MI lineage-reference JSON while a normal tagged checkout produces LF bytes; direct clean-tag execution therefore fails that raw input assertion until the historical CRLF representation is restored. The reference was not read for runtime configuration, so the defect does not change endpoint arithmetic, but it narrows replayability (`FIR_PROSPECTIVE_SW_V3_REPLAY_ERRATUM.md`).

The paired-difference sample SD was **0.000563779**. In registered seed order, the eight learned-minus-identity differences were **+0.005197548, +0.004308974, +0.005082181, +0.005788969, +0.005439290, +0.005516648, +0.004194543, and +0.004966712**. A post-hoc two-sided exact sign test for 8/8 positive differences gives $p = .0078125$. This sign test is a small-n robustness sensitivity, not the frozen decision rule and not population-level uncertainty across users, cutoffs, categories, or domains.

Figure 1 is a visual index of the released FIR contrasts, arranged to expose both scope and boundary conditions. It is not a pooled estimate: the intervals retain the estimators and evidence classes of their source adjudications and do not form one multiplicity family. The positive Amazon rows are outcome-known internal evidence; the matched MI controls show that learned FIR exceeds the equal-parameter pointwise placebo but not the shared causal filter; and the two prospectively frozen same-investigator MovieLens transfer intervals include zero.

Prospectively frozen non-Amazon replication and efficiency study (`PREREG_FIR_EFFICIENCY_ML1M_V1.md`; MovieLens 1M; verdict `ML1M-NO-FIR-REPLICATION`). The protocol and complete execution chain were committed and pushed before acquiring MovieLens. The rating-at-least-4 primary view used a global 90% time boundary, a training-observed catalog, 1,033 users, 2,359 items, and 135,131 training rows. Acquisition necessarily read and hash-bound TEST and used its target for cohort eligibility; the narrower sequestration claim is that model fitting/selection emitted no TEST scores and all 96 selected checkpoints existed before the 96 sealed one-shot TEST evaluations. Those evaluations preceded the protocol-designated adjudicator's first endpoint read. This is prospectively frozen same-investigator non-Amazon evidence, not independent confirmation or population-wide generalization.

The exact primary verdict is negative. Learned FIR did not replicate against identity (**+0.000000, ordinary paired 95% CI [-0.000074, +0.000075],** $p_{\text{Holm}} = .995$) or the equal-parameter pointwise arm (**+0.000035 [-0.000057, +0.000127],** $p_{\text{Holm}} = .796$). Mean NDCG@10 was 0.052151 for both learned and identity and 0.052116 for pointwise. The all-ratings sensitivity agreed: learned minus identity +0.000012 [-0.000107, +0.000131] and learned minus pointwise

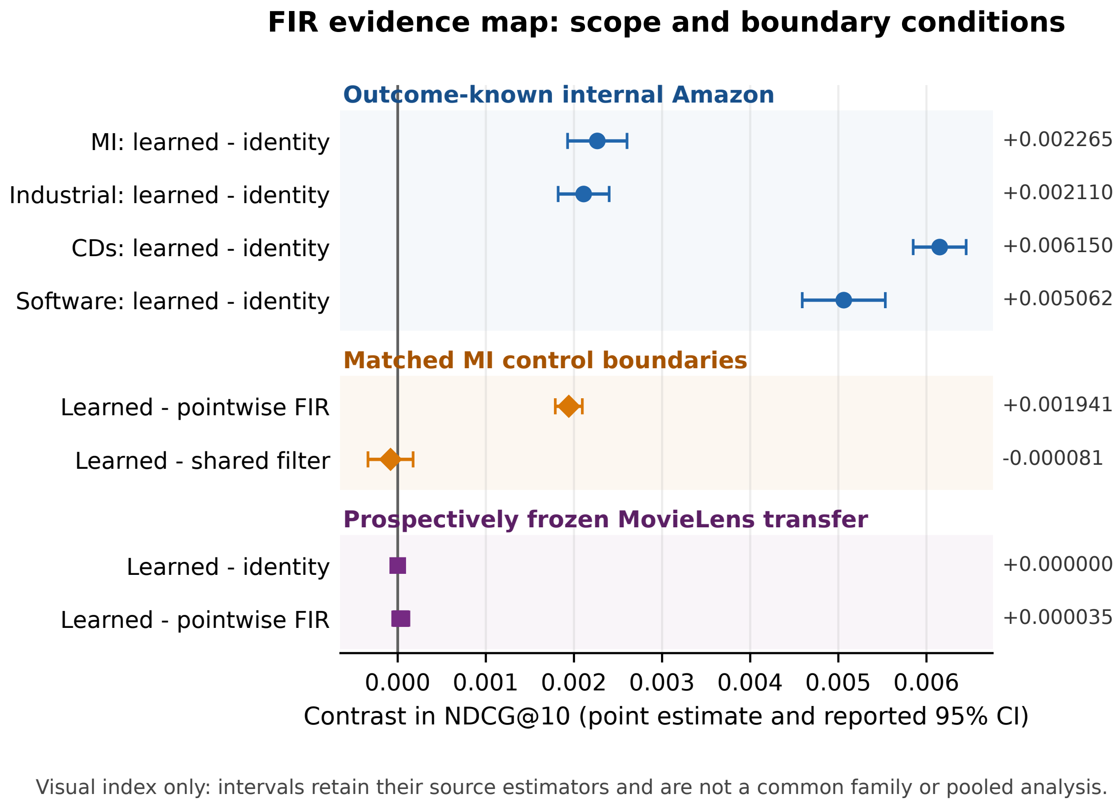


Fig. 1. FIR evidence map across released adjudications. Positive outcome-known Amazon contrasts coexist with a shared-filter boundary and a negative prospectively frozen MovieLens transfer result. Intervals retain their source estimators and evidence classes; they are not pooled and do not form one common multiplicity family. Exact values and source keys are in `figures/fig_fir_evidence_summary_data.csv`.

Table 3. MovieLens 1M R4 primary FIR replication and conditional parsimony. NI-PASS uses the frozen 0.000500 noninferiority margin versus learned FIR. Because the learned-FIR replication gate failed, NI-PASS is conditional coefficient-count compression only, not evidence of FIR utility or an end-to-end compute reduction.

arm	filter parameters	mean NDCG@10	arm − learned	simultaneous lower bound	frozen interpretation
identity	0	0.052151	−0.000000	—	learned replication failed
shared K=16	16	0.052211	+0.000060	−0.000068	NI-PASS
grouped K=16	128	0.052125	−0.000027	−0.000128	NI-PASS
low-rank K=16	320	0.052211	+0.000060	−0.000033	NI-PASS
learned per-channel K=16	1,024	0.052151	0	—	effect gate failed; no replicated FIR benefit
pointwise placebo	1,024	0.052116	−0.000035	—	learned−pointwise failed

+0.000043 [-0.000082, +0.000168]. All six pre-declared user/item cluster percentile intervals for the parsimonious-arm contrasts included zero.

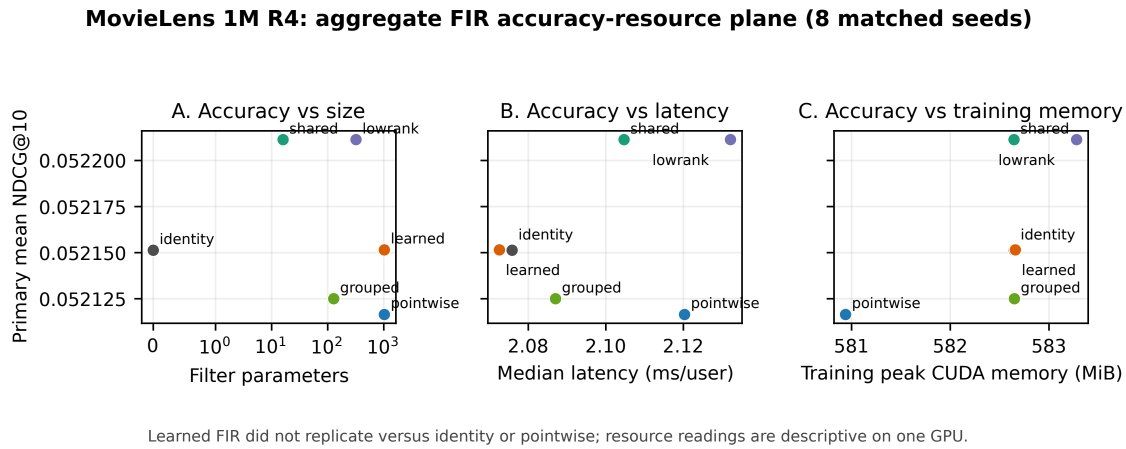


Fig. 2. MovieLens 1M R4 aggregate FIR accuracy-resource plane. Learned FIR failed to replicate versus identity and pointwise. Resource readings are descriptive on one GPU; exact aggregate values are in `figures/fig_fir_efficiency_ml1m_v1_data.csv`.

Shared, grouped, and low-rank passed the frozen noninferiority family ($p_{\text{Holm}} = 5.20 \times 10^{-6}$, 5.20×10^{-6}, and 1.43×10^{-6}). These are conditional coefficient-count compression findings, not a computational optimization and not evidence that a smaller FIR is useful, because the prerequisite learned-FIR replication contrasts failed. Median latency per user was 2.076/2.105/2.087/2.132/2.073/2.120 ms in table order; primary-view training peak memory ranged from 580.94 to 583.28 MiB on one RTX 5060 Ti. The resource plane is descriptive hardware evidence, not a general efficiency ranking. Under the ML-1M README, record-level rows, checkpoints, endpoints, and per-user sidecars remain private; the public graph verifies frozen code hashes and recomputes all reported aggregate-vector arithmetic but cannot independently replay private endpoint extraction.

Pre-declared breadth (two further categories). Under `PREREG_FIR_BREADTH.md` (committed before any run; fresh never-inspected seeds 20260713–17; mechanical adjudication in `FIR_BREADTH_RESULTS.md`), the filter was transplanted with **zero per-category tuning** and tested as a same-seed 5-seed filter-vs-no-filter contrast — an internal contrast with no comparator involved — on two categories never used in its development. **Both categories fired the frozen decision rule**: Industrial_and_Scientific same-seed Δ = **+0.0024 ± 0.0005** (pre-declared paired-analysis 95% CI [+0.0018, +0.0030]; 5/5 same-seed differences positive) and CDs_and_Vinyl same-seed Δ = **+0.0057 ± 0.0006** (pre-declared paired-analysis 95% CI [+0.0049, +0.0064]; 5/5). **Analysis-premise correction (2026-07-19):** the frozen rule *as written* is a paired Student-t, but its pairing premise is now known to be false — same-numbered seeds are not initialization-paired (§5.3 randomization disclosure) — so the pre-declared intervals are reported as executed-as-frozen outputs whose *paired interpretation is withdrawn*; the **primary supported statement** is the post-hoc independent-arm Welch analysis (we no longer call it conservative: with correlated arms neither interval has a general conservative ordering, and here the Welch intervals are in fact narrower than the paired ones), under which both effects remain positive with 95% CIs excluding zero (Industrial_and_Scientific [+0.0019, +0.0029], CDs_and_Vinyl [+0.0050, +0.0063]; graphed as `firb.*.welch`, evidence-class exploratory since the Welch analysis is post-hoc). Per the frozen claim wording as narrowed by this correction, each result is exactly: *the causal-FIR-filter arm's 5-seed improvement over the no-filter arm on that category is positive with a 95% CI excluding zero* — an internal same-seed contrast of the

FIR-plus-initialization/optimizer package; nothing broader, no SOTA language, no comparator statement. The filter package is thus multi-seed-supported on **four categories**, two of them under this pre-declared breadth campaign — and now **estimated under the valid independent-arm TFV2 pre-declaration (outcome-visible; not confirmatory — §5.3(vii))** (below), which supersedes this campaign's withdrawn paired interpretation as the primary FIR evidence.

TFV2 independent-arm replication (pre-declared, outcome-visible; adjudicated 2026-07-20). The `PREREG_TAIL_FIR_V2.md` campaign (Git-committed frozen rules; outcome-visible — OTS chronology in §5.3, disclosure (vii)) reran both breadth categories as fully independent 8-vs-8 arms (fresh never-inspected seeds; no pairing premise; per-run sidecars released): Industrial_and_Scientific overall Δ = **+0.002131** (t = 16.99, 95% CI [+0.001862, +0.002400]) and CDs_and_Vinyl overall Δ = **+0.005770** (t = 26.12, 95% CI [+0.005275, +0.006266]) — **both PASS under Holm** (one family with the tail endpoint E1, §5.3). This replaces the withdrawn paired interpretation with a valid pre-declared independent-arm design; the treatment remains the FIR-plus-initialization/optimizer package. Descriptively, the filter's positive-frequency-tail improvements are +0.000775 (IS, $p = 2\times10^{-4}$) and +0.002329 (CDs, $p = 3\times10^{-12}$).

Comparator ablations (audit-requested). Each arm targets one named design element of the filter (with the disclosed caveat that the singular headline initialization means these arms also alter the optimization path, §3) on the same 5-seed MI stack: freezing the kernel at a causal moving average (learnable gate retained) recovers only +0.00133 of the filter's +0.00225 descriptive mean gain over the no-filter baseline (0.04046 ± 0.00022 vs 0.04138 ± 0.00053) — an observation confounded by the shared singular start (§3). Here ± denotes sample standard deviation, and no inference is based on visual overlap of those bands; no learned-shape attribution is claimed. Freezing the gate at 1 (delta-initialized kernel) has a similar descriptive mean to the full filter (0.04127 ± 0.00046), so the zero-init gate comparison is confounded by the singular-initialization bootstrap (§3 disclosure) and is not a clean single-factor attribution.

Pre-declared confirmation vs the published per-category reference. The seeds above (20260608–17) are *exploratory*: kernel choice, epoch budget, and the decision to compare against the published Musical_Instruments HSTU-BLaIR number (0.0406; Liu 2025, Table 2) were all made while inspecting them. To make the comparison selection-free, we froze a version-controlled pre-declared protocol (`SOTA_CONFIRM_PREREG_V2.md`, committed to version control before any confirmation run: exact config and commands, data SHA256s, decision rule, claim wording) and ran **five pre-specified never-inspected seeds (20260618–22) under each of the two kernels — ten arm-runs** with per-run provenance manifests (command, git commit + clean-tree flag, environment, data hashes) and per-user record sidecars. Result: **K=16 fresh 5-seed 0.04152 ± 0.00045 (95% CI lower bound 0.04096) and K=8 0.04120 ± 0.00030 (CI-LB 0.04083) — both CI lower bounds above the published 0.0406, 10/10 seeds above.** The supported claim is exactly this: *fresh multi-seed means and seed-level confidence intervals exceed the published HSTU-BLaIR point estimate on this category under our reproduced protocol (parity evidence: users/items match the paper exactly; interactions differ by one — see* `SOTA_CONFIRM_PREREG_V2_ERRATA.md` *E1; our plain SASRec floor is* below *their published SASRec, ruling out an easier split).* The comparator is single-seed; our post-hoc local regeneration of it (§5.3 — the reference implementation's research path under data-movement shims, unpinned environment) regenerates the published value at its best full-eval epoch (0.0406; final epoch 0.0391) but is itself a single environment-caveated run, so no paired or distributional superiority is claimed, and this remains a per-category result — Video_Games is explicitly not claimed (§5.1). The first execution of this confirmation was voided by its own provenance tripwire (a concurrent documentation edit dirtied the tracked tree mid-campaign) and re-executed in full under a clean tree; both executions agree per-seed to ±0.0003 (`SOTA_CONFIRM_V2_RESULTS.md`). One protocol deviation is disclosed rather than claimed away: the gated runs executed at a documentation-only descendant commit of the pre-declaration's introducing commit (the pre-declared commit-equality rule was not literally satisfied); code identity across the two commits is demonstrated by empty protocol-file diffs and, in the from-scratch rebuild, by code hashes embedded in each run manifest (`SOTA_CONFIRM_PREREG_V2_ERRATA.md`, E3).

§5.2 (regenerated): pre-declared dual-kernel V2 confirmation (SOTACONF_V2, fresh seeds 20260618–22, EXEC2 gated artifacts, n_eval=57,439).

kernel	fresh 5-seed NDCG@10	95% CI lower bound	vs published 0.0406
K=16	0.04152 ± 0.00045	0.04096	ABOVE
K=8	0.04120 ± 0.00030	0.04083	ABOVE

Fresh seeds above 0.0406: 10/10. EXEC1↔EXEC2 max per-seed $|\Delta|$ = 0.000057 (< 0.0003). Clean rebuild (rebuild_v2/): K=16 CI-LB 0.04092, K=8 CI-LB 0.04084 — dual gate PASS.

Office_Products V1 (the original second-category attempt) — descriptive only; superseded by the counted V3 campaign below. Office_Products numerically exceeded the published HSTU-BLaIR point estimate under the frozen MI configuration, but the pre-declared SASRec floor check failed because our local SASRec floor was substantially above the published SASRec reference. We therefore treat the V1 Office campaign as descriptive evidence, not as a passed confirmatory category (the separately pre-declared V3 campaign later passed and is counted, §5.2). The floor anomaly has since been resolved mechanistically — the reference implementation's own SASRec, run locally on its own pipeline (§5.3), lands +13.9% above its published row, while a completed run of their Office HSTU-BLaIR configuration regenerates its published row (+1.6%; §5.3) — the conservatism is a property of the published Office SASRec row specifically. The VOID is retained on procedural grounds (the pre-declared floor check failed as written); full detail, decomposition, and disclosures in Appendix A.0.

Office_Products V3 (redesigned pre-declaration) — PASSED. Under `PREREG_OFFICE_V3.md` (committed before any run; fresh never-inspected seeds 20260728–32; gate: each arm's fresh 5-seed 95% CI lower bound must exceed the **environment-matched local regeneration of the comparator** — best full-eval 0.0279, itself above the published 0.0271 — with ≥4/5 seeds individually above), **both kernel arms passed**: K=16 fresh 5-seed **0.03047 ± 0.00011 (CI-LB 0.03033)** and K=8 **0.03029 ± 0.00005 (CI-LB 0.03024)**, 10/10 seeds above both references; dataset identity, reference-artifact hashes, and per-run provenance verified mechanically (`OFFICE_V3_RESULTS.md`). Per the frozen claim wording: *a per-category point-estimate comparison against the published 0.0271 and its environment-matched single-run local regeneration 0.0279; no paired or distributional superiority is claimed; this is not SOTA on Office_Products, not SOTA on Amazon Reviews 2023, and not a general-SOTA claim of any kind.* The V1 VOID above stands unchanged as its own record (Appendix A.0).

5.3 Secondary-study boundary

The remaining studies are supporting falsification and provenance evidence, not the inferential basis for the FIR claim. In the outcome-visible TFV2 text-versus-ID study, the Musical_Instruments positive-frequency tail contrast was +0.000420 [+0.000181,+0.000660], while Video_Games (+0.000173, p = .14) and the MI–VG interaction (+0.000247, p = .13) were not detected. Zero-training-exposure targets produced no hits through rank 100. Fixed-draw interaction/user thinning did not identify a mechanism: the most favorable corrected difference-in-differences was suggestive only (p = .058). These results support neither cold-start capability nor a general cross-dataset tail regularity.

The capacity-adding probe screen is mostly single-seed and supports no powered exclusion; its five-seed connection-gate interval includes zero and checkpoint cadence was unequal. Local HSTU-BLaIR executions are environment-caveated single-run regenerations, not pinned reproductions or distributional comparisons. A separately pre-declared EASE late-fusion study yielded positive paired intervals on three categories, but it is a standard two-scorer system addressing a different question and is not FIR evidence. The five-category semantic reranker is outcome-visible and protocol-deviated: its gains occur at training frequency 1–5, frequency-zero targets do not change at rank 10, mid/head means fall, and overall changes are near zero. It is retained as a descriptive redistribution study without an architectural-novelty or cold-start claim.

Full tables, figures, chronology, and artifact identifiers for these supporting studies remain in the canonical reader edition and reviewer supplement. Omitting their extended narrative from the main TORS file changes no value, verdict, or claim boundary.

6 Discussion

6.1 Interpretation

The main result is deliberately modular and narrowly identified. Positive FIR–identity contrasts on three outcome-visible Amazon categories are joined by a positive, thresholded Software robustness contrast. They show improvement only in those tested Amazon settings: the prospectively frozen MovieLens 1M study did not replicate learned FIR against identity or pointwise. V3 improves the protocol chronology, but unverified non-visibility of earlier V2 validation output, same investigator/code lineage/Amazon family, and local custody keep it outcome-known and non-independent. An equal-parameter current-position-only DCT/GELU placebo does not show a detected improvement over identity, and learned FIR exceeds it under matched initialization. This discriminates learned FIR from that compound residual but does not isolate temporal access because basis/rank, activation, channel mixing, and temporal access change together. The shared causal filter is also not separated from learned per-channel taps. On MovieLens, shared, grouped, and low-rank FIR arms met the noninferiority margin against learned FIR, but the learned-FIR effect gate failed; parsimony is therefore conditional coefficient-count compression rather than evidence of a useful or computationally cheaper module. No bypass implementation was tested, and measured end-to-end latency and memory did not materially improve. The official-code WEARec feasibility run also scored below our existing full-model reference under the shared evaluator. The frozen AlphaFuse-style package scored above both its zero-initialized upstream-class SASRec-ID control and the later parser-default `Normal(0,1)` sensitivity, but below that same full-model reference. Normal initialization improved SASRec-ID relative to the earlier zero-initialized control. Together these outcomes improve current-baseline coverage while preserving the mixed ranking. The normal-init contrast is cross-campaign and confounds phase/date with initialization; none of these studies equalizes architecture/capacity/tuning, isolates null-space fusion or initialization, or supplies independent confirmation. The evidence establishes neither per-channel-tap necessity, a universal frequency-filter benefit, cross-domain generalization, nor a new end-to-end architecture.

The supporting text studies sharpen that boundary. Frozen text features add only +2.7% overall on Video_Games, and TAPE is sub-additive (+0.0009 in one seed; +0.0004 in the four-seed check). The repaired MI analysis finds a small frequency-5-heavy benefit rather than cold-start capability: zero-exposure targets receive no hits through rank 100, the exclude-boundary sensitivity is nonsignificant, and the MI–VG contrast does not replicate. In the Beauty_and_PC scan (Appendix A.1), the positive rows bundle BLaIR, richer text, and the MLP adaptor, while the single MiniLM Linear-to-MLP comparison shows no improvement. The scan therefore does not isolate whether encoder, content, projection, or their interactions account for the bundled difference.

6.2 Practical implications

The FIR residual is small, causal, and detachable: it can be initialized as the identity and added without changing the item scorer or evaluation protocol. MovieLens resource measurements show no practically meaningful accuracy-resource frontier for learned FIR in the tested setting; latency and peak-memory differences are small, hardware-specific, and subordinate to the failed effect gate. In this implementation, chunked full softmax makes 207k-item training feasible, cached item features substantially reduce evaluation cost, and right-padding with a causal-only mask avoids the NaN failure observed with the former masking path. Conversely, heavy dropout, removal of learned item embeddings, and simple subsequence augmentation did not transfer in the Beauty supporting study; these are observations under this pipeline, not general exclusions.

6.3 Claim limits and remaining tests

- **Comparator uncertainty.** The counted Musical_Instruments and Office V3 comparisons are against published single-run point estimates and an environment-matched single-run regeneration; no paired or distributional superiority over HSTU-BLaIR is claimed. The official end-to-end CUDA/Triton system cannot run on the local sm_120 GPU. Only the research block at an explicitly aligned configuration is numerically matched; trained-checkpoint and end-to-end equivalence are not claimed.
- **Historical, canonical, control, and frozen Software evidence.** Same-numbered historical arms were not initialization-paired, so their paired interpretations are withdrawn. E-A uses its frozen Welch/Satterthwaite analysis; canonical breadth uses matched configuration blocks, but both reuse outcome-visible splits. The six-arm final evaluation was sealed, but ran across a dirty evolving tree and lacked independent sidecar custody. Fixed MA/HP are algebraically redundant; shared and nonlinear controls prevent a learned-tap-specific claim. The pointwise study adds a learned-FIR-versus-compound-current-only contrast on outcome-known MI, not temporal isolation or per-channel necessity. Software V3 used validation-only selection and sealed final evaluation, but V2-output non-visibility is not independently established; we classify it outcome-known/exploratory. Its local seals are hash-linked rather than immutable external custody, and its frozen tag has a disclosed raw-line-ending reference-hash replay defect. Office V1 remains **VOID**; Office V3 supports only its frozen point-estimate wording.
- **Prospective MovieLens boundary.** The non-Amazon study improves selection chronology and TEST sequestration, but it is same-investigator and same-code-lineage on one dataset and one global-time split. Its negative verdict blocks a cross-domain FIR-benefit claim. Noninferiority of the parsimonious arms is conditional on the failed learned-FIR replication gate; it is not equivalence to identity, a deployment utility claim, or independent confirmation. User/item bootstrap intervals are fixed-dataset sensitivities. The public graph can recompute the released aggregate seed vectors but cannot independently replay non-redistributable MovieLens records or private endpoint files.
- **WEARec current-baseline boundary.** The official model/training code was run under the paper split, complete-history mask, full-catalog evaluator, cutoff, and tie rule after validation-only preset selection. The resulting `WEAREC-BELOW-EXISTING-REFERENCE` verdict is same-investigator evidence on an outcome-known

split. Architecture, loss, schedule, and tuning budgets differ; the unpaired Welch contrast is descriptive, and neither independent confirmation nor a SOTA claim follows. The graph replays NDCG vector arithmetic, not private endpoint extraction or unreleased HR/MRR vectors.

- **AlphaFuse-style current-comparator boundary.** E-E V3 froze the official upstream representation and SASRec classes, shared training configuration, complete-history-masked evaluator, and eight fresh optimizer seeds per arm before launch. MiniLM-384 replaces the published AlphaFuse text vectors. V3's positive package-minus-ID interval uses a zero-initialized upstream-class SASRec-ID control. E-E V4 separately tested eight fresh parser-default `Normal(0,1)` SASRec-ID runs; the package remained above that arm, and normal initialization improved SASRec-ID over the earlier zero-init arm. Official recipes may override initialization by dataset, so the parser setting is not a universal upstream default. V4 is cross-campaign: phase/date and initialization are confounded, same-numbered seeds are not paired, and architecture, text availability, trainable capacity, parameter allocation, and tuning remain unequal. The equal-budget factorial remains open. Negative intervals against the existing reference are reported with equal prominence. No contrast isolates null-space fusion or initialization, reproduces a published AlphaFuse table, establishes equal tuning, supplies independent confirmation, or permits a SOTA claim. User/item bootstrap intervals are fixed-dataset sensitivities; the public graph validates released aggregate arithmetic and ledger structure but cannot read private endpoints/sidecars or replay record-level resampling.
- **Tail and probe scope.** The tail result is AR2023-specific, small in absolute terms, concentrated at train frequency 5, and not replicated on a non-Amazon domain. The original cohorts mixed exposure regimes and split frequency ties; TFV2 repaired those estimands but was outcome-visible and is not classified as confirmatory. Most capacity probes are single-seed screens, and nonsignificance is never interpreted as equivalence.
- **Open controls.** Item-text permutation and objective/target-multiplicity parity controls remain unrun, so the small text-stack gain cannot yet be attributed entirely to semantic alignment. The thinning studies intervene synthetically on one fixed draw and do not identify the real-world data-generating process.
- **Provenance and input scope.** One confirmation campaign ran at a documentation-only descendant of its frozen commit; code hashes and empty protocol-file diffs support code identity, but the literal commit-equality deviation remains disclosed. The canonical model consumes only item IDs, timestamps, and frozen item-text embeddings—never outputs or representations from a comparator model.

7 Conclusion

This paper contributes a narrow causal-filter module and an evaluation discipline for testing it. The discipline combines git-frozen protocols, a fail-closed artifact graph, environment-caveated comparator regeneration, and symmetric adjudication. Its value is illustrated by the Office_Products V1 campaign: its own floor check VOIDed the comparison, and that verdict remains in the record even though a redesigned protocol later passed on fresh seeds.

The main empirical result is a family of identity-initialized, gradient-active, strictly left-causal residual modules. Learned FIR–identity estimates are positive on Musical_Instruments (+0.002265, 95% CI [0.001928, 0.002602]), Industrial_and_Scientific (+0.002110 [0.001820, 0.002399]), and CDs_and_Vinyl (+0.006150 [0.005849, 0.006450]), but these are outcome-visible internal estimates. A separately frozen Software robustness attempt found learned FIR minus identity +0.005062 [+0.004591, +0.005533], with 8/8 positive paired differences and the ordinary 95% CI lower bound above its pre-declared +0.000500 reporting threshold. Because earlier V2 validation-output non-visibility is not independently established and the work remains same-investigator, same-code-lineage, same-Amazon-family under local same-user custody, it is outcome-known robustness rather than independent confirmation. In the active-control study, every trainable left-causal arm improves identity; learned taps beat the redundant fixed-filter parameterizations but are not separated from the shared causal filter or parameter-matched nonlinear causal control. In the subsequent equal-parameter current-position-only placebo study, pointwise minus identity is -0.000069 [-0.000200, +0.000061], while learned FIR minus pointwise is +0.001941 [+0.001788, +0.002095]. This discriminates learned FIR from that compound current-only residual in outcome-known MI, but does not isolate temporal access. It does not establish per-channel-tap necessity. In the prospectively frozen MovieLens 1M study, however, learned FIR did not replicate against identity (+0.000000 [-0.000074, +0.000075], $p_{\mathrm{Holm}} = .995$) or pointwise (+0.000035 [-0.000057, +0.000127], $p_{\mathrm{Holm}} = .796$). Shared, grouped, and low-rank arms met the pre-declared noninferiority margin against learned FIR only as coefficient-count compression: the learned-FIR effect gate failed, and no compute reduction was observed. The evidence therefore supports neither unique necessity of learned per-channel FIR coefficients, a general FIR benefit, superiority over the stronger external reference, nor independent cross-category confirmation.

The official-code WEARec current-baseline campaign reached mean NDCG@10 0.059184 [0.058674, 0.059693], below the existing six-seed full-model reference 0.067337 [0.067063, 0.067611]; the descriptive unpaired contrast was -0.008154 [-0.008689, -0.007618]. This closes only the narrow official-code/equal-evaluation feasibility check. It does not equalize architectures, losses, schedules, or tuning budgets and is not independent confirmation or SOTA evidence.

The frozen AlphaFuse-style current-comparator campaign reached NDCG@10 0.048273 [0.048129, 0.048416], above a zero-initialized upstream-class SASRec-ID control at 0.039024 [0.038106, 0.039941] by +0.009249 [+0.008329, +0.010169], but below the existing full-model reference by -0.019065 [-0.019347, -0.018783]. This countable mixed result supports the value of the tested whole representation package over that ID backbone; it does not isolate null-space fusion, equalize architecture/capacity/tuning, reproduce AlphaFuse's published text setting, or provide independent confirmation or SOTA evidence.

A separately frozen parser-default `Normal(0,1)` SASRec-ID sensitivity reached 0.043065 [0.042645, 0.043486]. The AlphaFuse-style package remained above it by +0.005207 [+0.004779, +0.005635], while normal initialization improved SASRec-ID over the earlier zero-init arm by +0.004042 [+0.003089, +0.004995]. Because this comparison is outcome-known, same-investigator, and cross-campaign, phase/date and initialization are confounded. It closes the narrow zero-init sensitivity concern, not initialization isolation, equal-budget comparison, independent confirmation, or SOTA.

The secondary result is narrower. Frozen text features improve the repaired Musical_Instruments frequency-tail endpoint (+0.000420 [0.000181, 0.000660]), but the gain concentrates at train frequency 5; removing that boundary group leaves no detected effect. The corresponding cross-dataset contrast is nonsignificant, and thinning interventions do not identify a tail mechanism. We therefore report a dataset-specific frequency-5 case, not a general cross-dataset result. TAPE, late fusion, sparse-warm redistribution, and the probe screen are supporting or descriptive studies, not additional headline contributions.

The evidence has important limits: uncertainty is over optimizer seeds on reused fixed splits; comparator uncertainty is not distributionally matched; several campaigns are outcome-visible or protocol-deviated; and no item-text permutation control, temporal split replication, or deployment study has been completed. The defensible contribution is thus modular rather than architectural: a small causal FIR realization, its measured internal effect under the stated protocol, and an auditable record that preserves nulls, deviations, and VOID outcomes alongside positive results.

8 Code and Data Availability

Code, frozen protocols, result artifacts, and build scripts are public at https://github.com/Ray0419/bestrec-sota-results. The tracked `RELEASE_MANIFEST.json` binds source code, split derivatives, text caches, result JSONs, released per-user sidecars, and the pinned HSTU-BLaIR submodule by SHA-256. Running `bootstrap_public_clone.py` fetches and verifies release-class assets; running `rebuild_hstu_submission.py --strict` then checks HSTU parity, recomputes the 200 artifact-gated cells, verifies the manifest, and executes every governed adjudicator.

The numerical graph and release manifest have different scopes: every declared empirical cell is graph-bound, but not every transitive graph source is separately enumerated in the release manifest. Table 0's literature attribution remains citation-checked authored prose, but its quantitative FIR fields are generated from active graph cells and the strict rebuild verifies the generated region. Table 0 is not a separate numerical graph family. We make no broader "every printed claim is hash-manifested" claim.

MovieLens 1M record rows, transformed splits, checkpoints, endpoint files, and per-user sidecars are not redistributed under the ML-1M README. The public release contains the frozen acquisition/training/evaluation code, source and split hashes, aggregate seed vectors/statistics, and adjudication. Consequently, the public graph recomputes the aggregate MovieLens claims but cannot independently replay the private record-level endpoint extraction.

The WEARec release contains the frozen official-code adapter, validation-only selection record, eight-seed aggregate adjudication, reference-vector identities, and private endpoint hashes. Its graph cell independently recomputes the released NDCG summaries and descriptive Welch arithmetic. Checkpoints, sealed endpoint files, and per-user sidecars remain private, and HR/MRR raw-vector arithmetic is therefore not publicly replayed.

The public `v0.9-audit-evidence` release is the working evidence store. Historical campaigns for which checkpoints or per-user records were not retained are labeled as such and require regeneration for rank-level reconstruction. Raw Amazon Reviews 2023 archives are not redistributed; derived splits are tied to their public source and preprocessing code by hashes. Local-only audit sidecars are outside every printed claim and are precommitted by digest for reviewer access. A new archival deposit will be cut only after the manuscript, author metadata, licenses, and clean-clone build are finalized; the older `v1.1.11-deposit` tag is retained as a historical boundary, not represented as current.

9 Acknowledgments

We thank the Amazon Reviews 2023 maintainers [Hou et al. 2024] for releasing the dataset and accompanying reference implementations. The MLP adaptor design used in our best variant is from their published SASRecText configuration. We also acknowledge the GroupLens Research Group for making MovieLens 1M available for research and cite its dataset history [Harper and Konstan 2015]; no endorsement by GroupLens or the University of Minnesota is implied.

10 Ethics and Data Governance

This work uses two pre-existing research datasets: **Amazon Reviews 2023** [Hou et al. 2024] (McAuley Lab) and **MovieLens 1M** [Harper and Konstan 2015] (GroupLens). For Amazon Reviews 2023, the maintainers state publicly that they are not in a position to assign a license or dictate usage terms; **that statement is not an affirmative permission grant, and we do not treat it as one**. We redistribute derived Amazon interaction-split CSVs (never the raw dataset) with attribution on the basis of the dataset's public research availability, will remove them on maintainer, platform, or venue request, and flag the redistribution basis for venue-level review rather than asserting a legal right. Scope of Amazon data actually consumed: interaction tuples (pseudonymous user identifiers — remapped to dense integers by our preprocessing — item identifiers, and timestamps) and **item metadata text** (titles, categories, and store/seller name — the AR2023 `store` field; an earlier label said "brand", corrected 2026-07-20: the frozen cache text template's literal `brand:` prefix is retained as-is inside the cached strings and documented as a misnomer, because regenerating the caches would invalidate every frozen-text result) for the frozen text encoders. We do **not** process review text bodies, ratings-as-text, or product images, and we make no attempt at re-identification of any user. Released artifacts follow the same boundary: per-user evaluation sidecars contain only remapped integer IDs, target item IDs, and rank outcomes; the raw dataset is **not redistributed** (our releases carry SHA256 hashes and regeneration scripts instead, `RELEASE_MANIFEST.json`).

For MovieLens 1M, acquisition downloaded the official ZIP and retained it only in private local storage; the study parsed **only `ratings.dat`** (`UserID`, `MovieID`, whole-star `Rating`, and `Timestamp`) and did not consume `users.dat` demographics or `movies.dat` metadata. The primary view thresholded ratings at 4 and an all-ratings sensitivity was also reported. In accordance with the MovieLens 1M README, neither the archive, transformed record rows, split files, checkpoints, endpoints, nor per-user sidecars are redistributed. The public release contains acquisition/training/evaluation code, source and split hashes, aggregate seed vectors/statistics, and the adjudication only. MovieLens data are used solely for non-commercial research; no GroupLens or University of Minnesota endorsement is stated or implied. The maintainer must review institutional retention, deletion, access-control, and legal requirements before final deposit; technical hash controls do not substitute for that review.

No new data was collected and no interaction with human subjects took place. Amazon Reviews 2023 is publicly accessible, platform-pseudonymized product-review data; MovieLens 1M is separately governed ratings data subject to the restrictions stated above. No institutional review determination was sought or is claimed, and we do not infer any jurisdiction's requirements; ACM policy places responsibility for applicable institutional, ethical, and legal compliance on authors, and we accept it, with this section's factual basis available as supporting documentation on request. **Released Amazon sidecar identifier surface, stated exactly:** those per-user sidecars carry our preprocessing's dense remapped user indices (`user_id` 0, 1, 2, ...), a target-item index, and rank outcomes — not hashes. The dense indices are deterministically linkable to the platform's pseudonymous identifiers through the released Amazon split CSVs, so the sidecars are exactly as pseudonymous as that public dataset, no more; MovieLens per-user sidecars are not released. Released Amazon derivatives are retained under the tagged GitHub release; requests to remove affected Amazon assets will be handled by deleting them and regenerating the release manifest. MovieLens retention and deletion remain governed by its separate private-data review above. Finally, this is an offline evaluation study: recommender systems deployed on such data can amplify popularity and exposure biases — our long-tail analyses quantify one aspect of that concern — and nothing in this paper constitutes a deployment claim.

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